

Career Dynamics under Random Access Shocks: Evidence from Lucky Loser Allocations*

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Abstract

Does a single randomly assigned tournament opportunity persistently shift a professional athlete’s career trajectory? I study lucky losers in professional tennis—players who lose in qualifying but enter the main draw when another player withdraws. At Grand Slams, lucky losers are selected by lottery. In the full ATP Grand Slam sample ($N = 248$), a stacked specification shows that selection increases ranking points by 31.9 at 4 weeks ($p = 0.028$), 38.5 at 12 weeks ($p = 0.007$), and 49.2 at 26 weeks ($p = 0.054$). On the WTA tour ($N = 132$), effects are larger at short horizons—64.4 points at 4 weeks ($p = 0.003$) and 78.7 at 8 weeks ($p < 0.001$)—but fade by 26 weeks. Elo ratings are unchanged at every horizon on both tours. A match-level tournament performance model finds no significant effect on match win probability on either tour ($\hat{\delta} = -0.123$, $p = 0.119$; 8,543 ATP matches; $\hat{\delta} = +0.034$, $p = 0.760$; 5,391 WTA matches). Non-Grand-Slam estimates using a control function approach corroborate these patterns: ATP ranking point gains of 12.2 at 4 weeks ($p = 0.065$) and 28.1 at 26 weeks ($p = 0.009$). Access begets access, but not ability.

JEL Codes: J24, J44, L83

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*I thank seminar participants at Emory University for helpful comments. Replication code and data are available at <https://github.com/hugosantanna/tennis-11>. All errors are my own.

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1 Introduction

Do small, quasi-random access shocks affect career trajectories in selective labor markets? In tournament-style markets—where ranking determines access and access determines the opportunity to earn further ranking—a single exogenous opportunity may propagate through the system. This paper studies lucky losers in professional tennis: players who lose in the final round of qualifying but enter the main draw when another player withdraws. At Grand Slam tournaments, lucky losers are selected by lottery among the top four ranked qualifying losers, creating a natural experiment in which otherwise similar players are randomly granted or denied a main draw opportunity. The setting allows clean measurement of both tournament access and playing ability, enabling a separation that is rarely possible in other labor markets.

The mechanism behind career propagation in this setting is path dependence through the ranking system. Professional tennis uses a rolling 52-week ranking that determines direct entry into tournament main draws. A player who earns ranking points at one event may gain direct entry to subsequent events, generating further opportunities to earn points. This creates a feedback loop: access generates exposure, exposure generates accumulation, and accumulation generates future access. A one-time exogenous shock to ranking points can trigger this cascade—but only if the initial shock is large enough. Players who receive a lucky loser slot but lose in the first round earn minimal ranking points and trigger no cascade. Players who win one or more main draw matches earn enough points to cross direct-entry thresholds at lower-tier events, setting off a sequence of additional opportunities. The ranking system, however, does not adjust for opponent quality: a first-round win at a Grand Slam earns the same points regardless of the opponent’s strength. Whether this ranking inflation translates into worse match outcomes—an “overseeding” effect—is an empirical question.

The empirical strategy has three components. First, Grand Slam lucky loser allocation is random within the top-four pool, providing the primary identification. I estimate immediate effects at the focal event and dynamic outcomes at horizons $h \in \{4, 8, 12, 26, 52\}$ weeks, tracing the trajectory of access and ability from the immediate aftermath through one

year. Second, I extend the analysis to non-Grand-Slam events using a control function approach, where the generalized residual is computed from the exact enumeration probability that a player’s rank among qualifying losers falls within the top slots—a probability determined by the full combinatorial structure of qualifying match pairings. Third, a match-level tournament performance model estimates whether lucky loser entry changes subsequent win probability conditional on opponent strength, exploiting the Grand Slam lottery for identification.

In the full ATP Grand Slam sample ($N = 248$, with 132 treated and 116 controls across 62 events), the event fixed effects specification shows that lottery selection increases the probability of winning a main draw match by 32 percentage points ($p < 0.001$), adds 1.48 matches played ($p < 0.001$), and generates 29.66 ranking points at the focal event ($p < 0.001$). On the WTA tour ($N = 132$, with 54 treated and 78 controls), the immediate effects are similar: a 38 percentage point increase in winning a main draw match and 24.86 additional ranking points. Over the following weeks, ATP ranking points increase by 31.9 at 4 weeks ($p = 0.028$), 38.5 at 12 weeks ($p = 0.007$), and 49.2 at 26 weeks ($p = 0.054$), fading by 52 weeks as the rolling window replaces the initial gain. WTA effects are larger at short horizons—64.4 points at 4 weeks ($p = 0.003$) and 78.7 at 8 weeks ($p < 0.001$)—but dissipate faster, consistent with the denser WTA tournament calendar. Elo ratings—which adjust for opponent quality—are unchanged at every horizon on both tours. A match-level logit on the Grand Slam lottery sample finds no significant effect on match win probability on either tour: $\hat{\delta} = -0.123$ ($p = 0.119$; 8,543 matches from 130 players) on the ATP tour and $\hat{\delta} = +0.034$ ($p = 0.760$; 5,391 matches from 98 players) on the WTA tour. Ranking point gains come entirely from additional participation rather than improved match performance. The non-Grand-Slam control function estimates corroborate these patterns: ATP ranking point gains of 12.2 at 4 weeks ($p = 0.065$) and 28.1 at 26 weeks ($p = 0.009$), with significant gains in tournament access at longer horizons.

This paper makes three contributions. First, I provide the first causal estimates of how a

within-career opportunity shock affects trajectories in a tournament labor market. The initial conditions literature has established that graduating in a recession ([Oreopoulos et al., 2012](#)), initial job placement ([Oyer, 2006](#)), and the environment in which a person starts ([Chetty et al., 2014](#)) shape long-run outcomes. I test whether these forces operate for a marginal within-career shock in a setting where treatment is randomized and where both access and ability can be measured separately. Second, the access-versus-ability decomposition provides direct evidence on tournament theory: LL entry generates ranking point gains through additional participation, but match-level performance conditional on opponent strength is unchanged on both tours ($\hat{\delta} = -0.123$, $p = 0.119$ on the ATP tour; $\hat{\delta} = +0.034$, $p = 0.760$ on the WTA tour). This cleanly separates the access channel from the ability channel and shows that access alone drives the career effects. Third, [Maity et al. \(2025\)](#) study lucky losers using correlational methods and find positive associations between entry and career outcomes. I provide the first causal estimates using literal randomization (the Grand Slam lottery), confirming that LL entry generates real access benefits while leaving match-level ability unaffected.

The remainder of the paper proceeds as follows. Section [2](#) reviews the related literature. Section [3](#) describes the institutional setting and data. Section [4](#) presents the econometric framework and identification strategies. Section [5](#) reports the main results. Section [6](#) discusses mechanisms. Section [7](#) presents robustness checks. Section [8](#) concludes.

2 Related Literature

This paper connects to the literature on career dynamics and path dependence. [Oyer \(2006\)](#) shows that initial labor market conditions have persistent effects on the careers of economists, with initial placement determining long-run research productivity and job quality. [Oreopoulos et al. \(2012\)](#) finds that graduating during a recession reduces earnings for up to a decade. [Oyer \(2008\)](#) demonstrates that stock market conditions at the time of MBA graduation

determine whether graduates enter investment banking, with lasting income consequences. [Chetty et al. \(2014\)](#) documents geographic variation in intergenerational mobility, establishing that the environment in which a person starts shapes their long-run trajectory. The common thread is that initial conditions matter because they determine access to subsequent opportunities—a form of cumulative advantage ([Merton, 1968](#)). The present paper tests whether this logic extends to within-career opportunity shocks that are randomly assigned, and separates the access and ability dimensions of the response.

A second strand examines the returns to marginal opportunities and access shocks. The closest labor market analog to the lucky loser setting is [Pallais \(2014\)](#), who shows that random provision of a single work opportunity on oDesk leads to persistent gains in subsequent employment—a finding that mirrors the access-begets-access mechanism documented here. [Abel et al. \(2020\)](#) find that randomly assigned reference letters improve labor market outcomes for disadvantaged job seekers in South Africa. [Startz \(2021\)](#) demonstrates that random interviewer assignment affects employment outcomes, establishing that small procedural shocks at the point of access can have lasting consequences. In science, [Azoulay et al. \(2014\)](#) document a Matthew effect in which early publication success cascades into future productivity gains, and [Wang et al. \(2019\)](#) find that scientists who narrowly miss early-career grant funding subsequently outperform those who narrowly succeed. [Azoulay et al. \(2010\)](#) show that the death of a superstar scientist reduces the productivity of their collaborators, documenting how access to high-quality peers shapes output. College access at the admission threshold produces long-run earnings gains ([Zimmerman, 2014](#); [Hoekstra, 2009](#); [Goodman et al., 2017](#)). In sports, [Berger and Pope \(2011\)](#) show that NBA teams trailing at halftime are more likely to win, providing evidence on motivational responses to adversity. This paper contributes to this growing literature on access shocks by providing literal randomization and the ability to separately measure access and ability responses—a distinction that existing studies of opportunity provision cannot typically make.

A third strand uses professional sports as laboratories for testing economic theories. [Gau-](#)

[riot and Page \(2019\)](#) use close tennis matches to show that the ranking system overrewards luck relative to skill. [González-Díaz et al. \(2012\)](#) study whether tennis players perform better in high-stakes moments. [Sunde \(2009\)](#) tests tournament theory predictions about heterogeneity and incentives. [Ehrenberg and Bognanno \(1990\)](#) examines whether golf tournament prize structures have incentive effects. Most directly related, [Maity et al. \(2025\)](#) study lucky losers across professional sports and find that early-career setbacks are associated with stronger subsequent performance. Their analysis is correlational: they compare lucky losers to other qualifying losers without accounting for the selection that determines who receives a lucky loser slot. The present paper provides the first causal estimates by exploiting the Grand Slam lottery mechanism.

The gap this paper fills is the absence of clean causal identification combined with dynamic career tracking. Existing studies of initial conditions use quasi-experimental variation that is typically one-shot (graduating in a recession, narrowly winning an election) and measure outcomes at a single long-run horizon. The lucky loser setting provides literal randomization, dynamic outcomes at multiple horizons from 4 to 52 weeks, and the ability to separately measure access and ability. The finding that access effects persist while ability effects are null on the ATP tour—and marginally positive on the WTA tour—is a distinction that the existing literature cannot make because it typically observes only one dimension of career capital.

3 Institutional Setting and Data

3.1 Institutional Background

Professional tennis is organized into a hierarchical tournament structure by the Association of Tennis Professionals (ATP, men) and the Women’s Tennis Association (WTA, women). Four Grand Slam tournaments—the Australian Open, Roland Garros, Wimbledon, and the US Open—sit at the top of the hierarchy, awarding up to 2,000 ranking points for the

winner. Below them are Masters 1000 events, followed by ATP/WTa 500 and 250 events. Each tournament awards ranking points based on the round reached, with higher-tier events offering more points per round. Both tours use a rolling 52-week ranking: a player’s ranking equals the sum of points from their best results over the previous year, updated weekly. Rankings determine direct entry into tournament main draws: the top 104 players enter Grand Slam main draws directly, the top 40–56 enter Masters events, and similar thresholds apply at lower tiers. This structure creates a feedback loop central to the analysis: ranking points determine tournament access, and tournament access determines opportunities to earn ranking points.

Players ranked below the direct-entry threshold can enter a tournament’s qualifying draw, a mini-tournament held before the main event. Qualifying draws are single-elimination brackets—typically 32 or 16 players on the ATP tour, 12 or 16 on the WTA tour—with 4 to 8 qualifiers advancing to the main draw after winning two or three consecutive matches. The players who populate qualifying draws are ranked approximately 100th to 300th in the world: too low for direct entry but competitive enough to warrant a spot in the qualifying field. These marginal players are the focus of this study. The two tours differ in ways that matter for the analysis. The WTA calendar is denser, with more tournaments per month and shorter gaps between events. WTA qualifying draws are typically smaller (often 12 players compared to 16 on the ATP side), meaning fewer final-round qualifying losers per event.

When a player who has been accepted into a tournament main draw withdraws—due to injury, illness, or scheduling conflicts—the vacant slot is filled by a “lucky loser” (LL): a player who lost in the final round of qualifying. The selection mechanism differs by tournament tier, and this difference is central to the identification strategy. At Grand Slam tournaments, the rulebook specifies a two-track system based on withdrawal timing. When a main draw withdrawal occurs *before* qualifying is complete, the top four ranked final-round qualifying losers enter a random draw (lottery) for the available slot. Each of the four has an equal probability of selection regardless of ranking differences within the group. When a withdrawal

occurs *after* qualifying ends, the highest-ranked final-round qualifying loser receives the slot without a lottery. The lottery pathway commenced after Wimbledon 2005 and provides the basis for the primary identification strategy. At non-Grand-Slam events (Masters 1000, ATP 500, ATP 250), lucky loser selection follows a strict ranking rule: the highest-ranked final-round qualifying loser receives the first available slot, the second-highest receives the next, and so on. There is no lottery component.

3.2 Data Sources and Sample Construction

The primary data source is Jeff Sackmann’s Tennis Abstract database, a comprehensive public dataset of professional tennis match results covering the ATP tour from 1968 to 2024 and the WTA tour over the same period. The database includes match outcomes, player rankings, tournament details (surface, tier, draw size), and entry type (direct acceptance, qualifying, wildcard, or lucky loser). I supplement match records with weekly ranking files to construct ranking trajectories before and after each event. I compute Elo ratings from the full match history using a K -factor of 32 and an initial rating of 1,500. Elo ratings adjust for opponent quality: beating a higher-rated player produces a larger Elo gain than beating a lower-rated one. The distinction between ranking points (which do not adjust for opponent quality) and Elo ratings (which do) is central to the mechanism analysis.

The unit of observation is a player-episode pair (i, e) , where player i is a final-round qualifying loser at event e . The eligible pool consists of all final-round qualifying losers at Grand Slam events where at least one lucky loser slot was awarded, restricted to the top four ranked losers per event. Players at events with zero LL entries are excluded because the absence of main draw withdrawals means no player at that event faced any probability of receiving LL treatment. Grand Slam qualifying data with lottery-based assignment are available from 2006 to 2024.

The treatment variable is defined as:

$$D_{ie} = 1 \text{ if player } i \text{ entered the main draw as a lucky loser at event } e.$$

For the Grand Slam lottery design, treatment assignment is random within the top-four pool. For the non-Grand-Slam control function design, treatment is determined by the ranking rule, and endogeneity is corrected using a control function derived from the peer component of the selection probability.

The primary estimation sample consists of all top-four ranked final-round qualifying losers at Grand Slams from 2006 to 2024. The ATP Grand Slam sample contains $N = 248$ player-episode observations (132 treated, 116 control) drawn from 174 unique players across 62 events. The WTA Grand Slam sample contains $N = 132$ observations (54 treated, 78 control) from 112 unique players across 33 events. I also construct several restricted samples. The *first-LL-only* sample restricts to each player’s first appearance in the Grand Slam LL candidate pool, yielding $N = 174$ on the ATP tour (91 treated, 83 control) and $N = 112$ on the WTA tour (48 treated, 64 control). This restriction eliminates contamination from prior LL treatment. The *verified lottery subsample* identifies events where the selected LL was provably not the highest-ranked candidate, confirming that a lottery rather than a ranking rule was used: $N = 100$ on the ATP tour (38 treated, across 34 verified events) and $N = 44$ on the WTA tour (14 treated). The *non-Grand-Slam sample* contains $N = 2,708$ ATP observations (874 treated). The WTA non-Grand-Slam sample covers 2009–2024 (when the current WTA tier system was adopted) and contains 2,545 player-episode observations (755 treated) across WTA 1000, WTA 500, and WTA 250 events.¹

¹WTA 1000 events (formerly Premier Mandatory) correspond to ATP Masters 1000; WTA 500 (formerly Premier) to ATP 500; WTA 250 (formerly International) to ATP 250.

3.3 Lucky Loser Distribution

Table 1 reports the distribution of LL entries by Grand Slam and tour. The four Grand Slams differ in their LL rates, reflecting variation in withdrawal patterns across tournaments. Roland Garros has the highest LL rate on the ATP side, while the Australian Open has the lowest. These cross-tournament differences do not threaten identification because the lottery operates within each Grand Slam independently. The distribution reveals that LL entries are not rare events: across the 62 ATP Grand Slam events in the sample, 132 of 248 top-four pool members received LL entry, a rate of 53 percent. On the WTA side, 54 of 132 pool members (41 percent) received entry. The lower WTA rate reflects smaller qualifying draws and fewer withdrawal opportunities. The non-Grand-Slam distribution is reported in the appendix (Table 24).

Table 1: Lucky Loser Distribution by Grand Slam and Tour						
Grand Slam	ATP			WTA		
	LL	Control	Total	LL	Control	Total
Australian Open	27	33	60	4	8	12
Roland Garros	45	27	72	20	28	48
Wimbledon	35	29	64	17	23	40
US Open	25	27	52	13	19	32
Total	132	116	248	54	78	132

Notes. Candidates are top-four ranked final-round qualifying losers at Grand Slam events where at least one LL slot was awarded. LL entries are players who entered the main draw as lucky losers. LL Rate is LL entries divided by candidates. Sample period: 2006–2024.

Table 2 cross-tabulates the number of LL candidacies against the number of LL slots won, separately for ATP and WTA. Most players appear as LL candidates only once or twice; a smaller group of repeat candidates accumulates multiple opportunities and wins. The non-Grand-Slam equivalent is reported in the appendix (Table 25).

Table 2: Player LL Candidacies and Wins: Grand Slam

Opportunities	ATP				WTA			
	Won=0	Won=1	Won=2	Won=3+	Won=0	Won=1	Won=2	Won=3+
1	54	67	0	0	55	41	0	0
2	12	18	9	0	7	4	2	0
3	0	6	3	1	0	1	1	0
4	0	0	1	1	0	0	1	0
5+	0	0	0	2	0	0	0	0
Total	66	91	13	4	62	46	4	0

Notes. Each cell reports the number of unique players with the given combination of LL candidacies (rows) and LL wins (columns). Rows sum to the total number of unique players who were LL candidates at that frequency. Sample period: 2006–2024.

3.4 Variable Construction

Outcomes are organized into two categories. *Immediate event-level outcomes* measure the direct effect of the LL slot at the focal event: (1) whether the player won at least one main draw match, (2) the number of main draw matches played, and (3) ranking points earned at the event. *Post-episode dynamic outcomes* measure career trajectories at horizons $h \in \{4, 8, 12, 26, 52\}$ weeks following the event. Primary dynamic outcomes include: the number of main draw entries (MainDraws_{ie}^h), the number of matches played at ATP/WTA 250+ events (Matches250_{ie}^h), and the change in ranking points ($\Delta\text{Points}_{ie}^h$). Secondary outcomes include the change in Elo rating (ΔElo_{ie}^h). The Elo outcome is the key test of ability: if LL entry improved playing skill, Elo ratings would rise.

Pre-draw state variables enter as controls:

$$Z_{ie}^{pre} = \{\text{Points}_{ie}, \text{Points}_{ie}^2, \text{Elo}_{ie}/100, (\text{Elo}_{ie}/100)^2, \text{SurfElo}_{ie}/100, (\text{SurfElo}_{ie}/100)^2, \text{GS_LL_Won}_{ie}, \text{GS_}$$

Pre-treatment controls include ranking points (linear and quadratic), Elo rating / 100 (linear and quadratic), surface Elo / 100 (linear and quadratic), prior Grand Slam LL spots won, prior Grand Slam LL candidacies not won, prior non-Grand-Slam LL spots won, prior non-Grand-Slam LL candidacies not won, and player age. Elo ratings are divided by 100 when

used as covariates to improve coefficient scale (raw Elo ranges from approximately 1,400 to 2,200); Elo *outcomes* (change in Elo rating) remain in raw points for direct interpretability. Ranking points and Elo absorb the mechanical confound that higher-ranked LL candidates have better tournament access regardless of LL entry. The LL history variables control for differential exposure to prior LL opportunities, separating the effect of the current episode from accumulated treatment history. Age captures career stage effects. In pooled ATP–WTA specifications, a tour indicator is added.

A guiding design principle is that only pre-treatment variables are included as controls. Conditioning on post-treatment variables (e.g., contemporaneous Elo or ranking at horizon h) would absorb part of the causal pathway. Given the sample sizes—248 observations in the full ATP lottery sample, 132 in the WTA sample—I favor simpler models with fewer explanatory variables to preserve degrees of freedom.

3.5 Summary Statistics

Tables 3 and 4 report summary statistics for the ATP and WTA Grand Slam estimation samples, respectively.

Players in the Grand Slam lottery pool are ranked approximately 100th to 250th in the world, with a mean Elo rating around 1,700. Ages range from 18 to 35, with a mean of approximately 24 years. The distribution is right-skewed in ranking (a few players near the top 100 mixed with many ranked 150–250) and approximately symmetric in Elo.

The balance tests (reported in Section 7.4) show that the ATP sample has a joint F -test p -value of 0.230 and the WTA sample has a joint F -test p -value of 0.333. These results are consistent with randomization within the top-four pool.

The players in this sample are drawn from the same competitive margin: final-round qualifying losers ranked in the top four at their event. They are heterogeneous in age, ranking, and career stage, but the randomization (at Grand Slams) ensures that treatment and control groups are drawn from the same distribution. The non-Grand-Slam samples

have similar characteristics but are larger (ATP: $N = 2,708$; WTA: $N = 2,545$), providing greater statistical power at the cost of relying on the control function for identification. Non-Grand-Slam summary statistics are reported in the appendix (Tables 26 and 27).

Table 3: Summary Statistics: ATP Grand Slam Sample

Variable	LL Selected			Not Selected		
	Mean	SD	N	Mean	SD	N
<i>Panel A: Pre-Treatment</i>						
Age	26.89	4.00	132	27.08	3.20	116
Ranking points	492.36	108.57	132	468.99	105.09	116
Elo rating	1750.71	65.27	132	1745.65	62.05	116
Surface Elo	1652.53	123.70	132	1658.97	125.33	116
Prior GS LL won	0.30	0.68	132	0.25	0.49	116
Prior GS LL not won	1.11	1.44	132	0.88	1.06	116
Prior non-GS LL won	1.15	1.62	132	0.74	1.18	116
Prior non-GS LL not won	1.97	2.26	132	1.69	2.00	116
Had prior LL	0.54	0.50	132	0.49	0.50	116
<i>Panel B: Outcomes</i>						
Ranking points Δ (4w)	16.68	65.80	132	-4.17	50.28	116
Ranking points Δ (26w)	41.25	210.78	106	1.64	227.87	94
Ranking points Δ (52w)	22.33	295.30	87	34.57	315.17	79
Elo Δ (4w)	1.13	25.00	129	0.68	28.82	113
Elo Δ (26w)	7.34	64.95	89	1.97	56.39	88
Elo Δ (52w)	-0.56	78.18	77	9.94	83.55	70
Main draws entered (26w)	4.85	3.52	112	4.05	2.52	97
Matches at 250+ events (26w)	7.66	7.02	112	6.21	5.15	97

Notes. Top-four ranked final-round qualifying losers at ATP Grand Slam events where at least one LL slot was awarded, 2006–2024. $N = 248$ (132 treated, 116 control). All explanatory variables are measured pre-treatment. Elo rating is scaled by dividing by 100 when used as a covariate; Elo change outcomes remain in raw points.

Table 4: Summary Statistics: WTA Grand Slam Sample

Variable	LL Selected			Not Selected		
	Mean	SD	<i>N</i>	Mean	SD	<i>N</i>
<i>Panel A: Pre-Treatment</i>						
Age	24.53	4.17	54	24.16	3.42	78
Ranking points	559.76	155.98	54	476.03	156.01	78
Elo rating	1927.07	100.14	54	1888.36	98.84	78
Surface Elo	1743.57	191.55	54	1743.83	166.42	78
Prior GS LL won	0.15	0.36	54	0.09	0.33	78
Prior GS LL not won	0.39	0.74	54	0.37	0.54	78
Prior non-GS LL won	0.65	1.20	54	0.53	0.92	78
Prior non-GS LL not won	1.81	2.16	54	1.27	2.00	78
Had prior LL	0.39	0.49	54	0.37	0.49	78
<i>Panel B: Outcomes</i>						
Ranking points Δ (4w)	40.67	69.91	54	12.21	51.98	78
Ranking points Δ (26w)	57.69	237.49	49	88.64	234.13	69
Ranking points Δ (52w)	57.42	420.51	40	91.42	315.28	62
Elo Δ (4w)	5.72	25.56	51	1.00	21.68	76
Elo Δ (26w)	14.33	48.99	35	3.40	51.47	56
Elo Δ (52w)	14.75	84.14	34	2.90	70.20	60
Main draws entered (26w)	4.60	2.96	50	3.76	2.61	72
Matches at 250+ events (26w)	6.82	5.81	50	4.78	4.59	72

Notes. Top-four ranked final-round qualifying losers at WTA Grand Slam events where at least one LL slot was awarded, 2006–2024. $N = 132$ (54 treated, 78 control). All explanatory variables are measured pre-treatment. Elo rating is scaled by dividing by 100 when used as a covariate.

4 Econometric Framework and Identification

4.1 Grand Slam Identification Strategy

The Grand Slam lottery provides random assignment of lucky loser entry within the top-four pool. The identifying assumption is:

$$E[Y_{ie}(0) \mid D_{ie} = 1, e] = E[Y_{ie}(0) \mid D_{ie} = 0, e]$$

That is, among the top four ranked final-round qualifying losers at a given Grand Slam, lottery-selected and unselected players have the same expected untreated potential outcomes. This holds by design: within the pool, each player has an equal probability of selection regardless of ranking, ability, or other characteristics.

The randomization operates within each event’s top-four pool, not across players globally. A player ranked 120th at the Australian Open and a player ranked 180th at Roland Garros are not comparable by virtue of the lottery alone; they are in different pools. The event fixed effect α_e absorbs all between-pool variation, so the estimated effect compares lottery winners to lottery losers within the same event.

A caveat is that the lottery mechanism operates only when withdrawals occur before qualifying completes. Post-qualifying withdrawals follow the ranking rule, so the sample may include some ranking-based assignments. The verified lottery subsample (Section 7.1) addresses this contamination concern by restricting to events where the selected LL was provably not the highest-ranked candidate.

With only four players per pool and a binary treatment, finite-sample imbalance is expected even under true randomization. Across the full ATP Grand Slam sample, the joint F -test yields $p = 0.230$; on the WTA side, $p = 0.333$. To improve precision and absorb residual imbalance, I include pre-treatment controls in all dynamic specifications. I also report specifications without controls as a complement.

4.2 Immediate Effects

The event-level model estimates the immediate effect of receiving the LL slot at the focal Grand Slam:

$$Y_{ie}^{imm} = \alpha_e + \delta D_{ie} + \varepsilon_{ie} \quad (1)$$

where Y_{ie}^{imm} is one of three event-level outcomes: (i) an indicator for winning at least one main draw match, (ii) the number of main draw matches played, and (iii) ranking points earned at the event. The event fixed effect α_e is a Grand Slam $\times$ Year indicator. The coefficient δ measures the size of the initial treatment shock—the immediate dose of ranking points and competitive experience that the LL slot provides. This quantity sets the scale for interpreting the dynamic effects that follow. Control outcomes are mechanically zero because unselected players do not enter the main draw.

4.3 Post-Episode Dynamics

The dynamic model traces career outcomes at horizons $h \in \{4, 8, 12, 26, 52\}$ weeks:

$$Y_{ie,h} = \alpha_e + \alpha_h + \beta_h D_{ie} + \Gamma Z_{ie}^{pre} + \varepsilon_{ie,h} \quad (2)$$

where $Y_{ie,h}$ is the outcome for player i from event e measured at horizon h , α_e is an event fixed effect (Grand Slam $\times$ Year), α_h is a horizon fixed effect absorbing level differences across time horizons, Z_{ie}^{pre} is the vector of pre-treatment controls (ranking points / 1000 linear and quadratic, Elo / 100 linear and quadratic, surface Elo / 100 linear and quadratic, prior GS LL spots won, prior GS LL candidacies not won, prior non-GS LL spots won, prior non-GS LL candidacies not won, and player age) with horizon-invariant coefficients Γ , and $\varepsilon_{ie,h}$ is the error term. Standard errors are clustered at the player level. The coefficients on Z^{pre} do not vary by horizon because these controls absorb pre-treatment differences that are constant across the post-event trajectory. The horizon fixed effects are important for count

outcomes (main draws entered, matches played) where level differences across horizons are substantial; adding α_h increases the R^2 from 0.35 to 0.53 for main draw entries.

The coefficient β_h identifies the causal effect of lottery selection on the outcome at horizon h . The sequence $\{\beta_4, \beta_8, \beta_{12}, \beta_{26}, \beta_{52}\}$ traces the divergence between treated and control trajectories after the random shock. Immediate effects ($h = 0$) are captured by the event-level model in equation (1) and are excluded from the dynamic specification to avoid double-counting.

A player may appear in the lottery pool at multiple events. To avoid contamination from subsequent treatments, I censor each player-episode at the date of the player’s next final qualifying round loss at an LL-granting event of the same type:

$$h^* = \min(12 \text{ months, time to next same-type qualifying loss at an LL-granting event})$$

For Grand Slam episodes, this means censoring at the next Grand Slam qualifying loss where at least one LL slot was awarded. For non-Grand-Slam episodes, censoring occurs at the next non-Grand-Slam qualifying loss at an LL-granting event. Within this window, the only relevant treatment is D_{ie} from the originating event, preventing contamination from overlapping treatment spells. A player who appears in the LL candidate pool at multiple events generates multiple episodes, potentially with different treatment statuses. As a robustness check, I restrict the sample to each player’s first Grand Slam LL opportunity (Section 7.2).

4.4 Heterogeneous and Treatment Dose Effects

I estimate heterogeneous treatment effects by augmenting equation (2) with interactions between the treatment indicator D_{ie} and category dummies. For a categorical moderator with K groups (one base category omitted), the estimating equation is:

$$Y_{ie,h} = \alpha_e + \alpha_h + \beta_h D_{ie} + \sum_{k=2}^K \delta_{k,h} D_{ie} \times \mathbf{1}[G_{ie} = k] + \Gamma Z_{ie}^{pre} + \varepsilon_{ie,h} \quad (3)$$

where β_h captures the base-category treatment effect at horizon h and $\delta_{k,h}$ captures the *differential* effect for category k relative to the base. Since β_h is already indexed by horizon in equation (2), this specification inherits the horizon-specific treatment structure without requiring additional horizon interactions. I examine three dimensions: pre-treatment ranking (above versus below the tour-specific median, with below-median as base), age (below versus above 24, with younger as base), and prior LL experience (none versus any, with no prior experience as base).

To characterize how treatment effects vary with dose, I interact D_{ie} with dummies for the number of main draw matches won at the focal event: one win, and two or more wins, with zero wins (first-round loss) as the base. The dose variable is endogenous—players who win more matches are better or luckier on the day—so these estimates are descriptive, not causal. They answer a different question than the ITT: conditional on receiving the LL slot, how much does the career impact depend on what the player does with it? Standard errors are clustered at the player level throughout.

4.5 Non-Grand-Slam Control Function Strategy

At non-Grand-Slam events, LL slots go to the highest-ranked final-round qualifying losers. I use a control function approach to account for endogenous LL assignment, exploiting the peer component of the selection probability computed from the combinatorial structure of qualifying match pairings.

Selection probability. The selection probability P_i^{LL} is the probability that player i 's rank among final-round qualifying losers falls within the top c slots (where c is the number of LL positions available), conditional on player i having lost. Formally:

$$P_i^{LL} = \Pr(\text{rank among losers} \leq c \mid i \text{ loses})$$

This probability depends on the qualifying match outcomes of *other* players—specifically, on the probability that enough higher-ranked qualifiers also lose their matches. Construction proceeds in two steps. First, I estimate a weighted logit model of match outcomes on qualifying round matches at non-Grand-Slam LL-granting events, separately for each tour (ATP: $N = 9,775$; WTA: $N = 10,199$ qualifying matches). The match-level covariates X_{ijm} include ranking points difference / 1000 (linear and quadratic), Elo difference / 100 (linear and quadratic), surface Elo difference / 100 (linear and quadratic), Laplace-smoothed head-to-head win proportion, head-to-head encounter count, and age difference, along with event-level surface indicators (clay and grass). The logit is estimated by weighted MLE, with weights equal to the total points in play (sum of service points) in each match; qualifying round number serves as a fallback weight when service point data are unavailable. Estimating on all qualifying rounds—not just the final round—improves parameter precision by exploiting the full set of qualifying matches. The same logit specification is used for the selection probability, the tournament performance model, and the treatment dose analysis, ensuring a unified covariate structure across applications. This yields predicted match-winning probabilities $\hat{P}(a_m \rightarrow b_m \mid X, Z)$ for each final-round qualifying match m between players a_m and b_m . Second, given these predicted probabilities and the $N_e - 1$ other final-round qualifying matches (excluding player i ’s match, since i is conditioned on having lost), I enumerate all 2^{N_e-1} possible outcome combinations. Each combination has probability equal to the product of the individual match outcome probabilities from the logit (invoking conditional independence of match outcomes given observables). For each combination, the set of losers is determined and their rankings compared; P_i^{LL} is the sum of probabilities of all combinations in which i ’s rank falls within the top c among losers. Player i ’s own loss probability does not enter P_i^{LL} ; the enumeration uses only final-round matches, since the LL selection mechanism conditions on the final qualifying round structure. The resulting P_i^{LL} is systematically lower than a naive Bernoulli convolution that ignores the draw structure (mean 0.279 vs. 0.506 across the non-GS sample), because the

exact enumeration correctly conditions on which players are paired against each other. The key identifying assumption is that, conditional on player i 's characteristics and the fact that i lost, the qualifying match outcomes of other players affect i 's future career only through the lucky loser channel.

Control function approach. For all non-Grand-Slam specifications, I account for endogenous LL assignment using a control function approach. The generalized residual $\hat{v}_{ie}$ is computed analytically from P_i^{LL} :

$$\hat{v}_{ie} = D_{ie} \cdot \frac{\phi(\Phi^{-1}(P_i^{LL}))}{P_i^{LL}} - (1 - D_{ie}) \cdot \frac{\phi(\Phi^{-1}(P_i^{LL}))}{1 - P_i^{LL}}$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are the standard normal PDF and CDF, respectively. This residual captures the degree to which player i 's actual LL status deviates from what P_i^{LL} would predict under normality. It enters the outcome equation as an additional regressor:

$$Y_{ie,h} = \alpha_e + \alpha_h + \beta_h D_{ie} + \rho \hat{v}_{ie} + \Gamma Z_{ie}^{pre} + \varepsilon_{ie,h} \quad (4)$$

The coefficient $\hat{\rho}$ on $\hat{v}_{ie}$ tests for endogeneity of LL entry: if $\hat{\rho} \neq 0$, LL assignment is correlated with unobservables and the control function correction is needed. The ATP non-Grand-Slam sample contains $N = 2,708$ observations (874 treated); the WTA sample contains $N = 2,545$ observations (755 treated). Standard errors are clustered at the player level.

Control function for the match-level model. For the match-level tournament performance model at non-Grand-Slam events (Section 6.3), the same analytically computed generalized residual enters the outcome logit:

$$p_{itm} = \Lambda(\beta' X_{ijm} + \gamma' Z_{ie}^{pre} + \delta \cdot D_{ie} + \rho \cdot \hat{v}_{ie}) \quad (5)$$

where X_{ijm} includes ranking points difference / 1000 (linear and quadratic), Elo difference / 100 (linear and quadratic), surface Elo difference / 100 (linear and quadratic), Laplace-smoothed H2H win proportion, H2H encounter count, age difference, and surface indicators (clay and grass)—the same logit specification used for the selection probability. Z_{ie}^{pre} includes ranking points / 1000 (linear and quadratic), Elo / 100 (linear and quadratic), surface Elo / 100 (linear and quadratic), and player age. In the dose specification, $\hat{v}_{ie}$ is interacted with the centered matches-won variable to allow the endogeneity correction to vary with treatment dose, because players who win more matches at the LL event may have different unobserved ability profiles. Standard errors are obtained via player-level block bootstrap (200 replications) to account for the generated regressor.

5 Results

All Grand Slam results are presented separately for the ATP and WTA tours. Non-Grand-Slam control function results are discussed in Section 5.5 and reported in full in the appendix.

5.1 Immediate Effects

Before examining post-episode dynamics, I characterize the treatment dose—the direct consequences of LL entry at the focal event. Because control players by definition do not enter the main draw, these estimates describe the magnitude of the treatment rather than a causal effect relative to a counterfactual.

Table 5 reports the immediate event-level effects of lottery selection.

On the ATP tour ($N = 248$), 32 percent of treated players win at least one main draw match, earning a median of approximately 35 additional ranking points at the event. Treated players play 1.48 additional matches ($p < 0.001$) and earn 29.66 ranking points on average ($p < 0.001$). The magnitude of the immediate shock is economically meaningful: a Grand Slam first-round loss earns 10 ranking points, while a second-round loss earns 45 points.

On the WTA tour ($N = 132$), 38 percent of treated players win at least one main draw match, play 1.41 additional matches ($p < 0.001$), and earn 24.86 ranking points on average ($p < 0.001$). The smaller WTA point total reflects the different competitive density at the margin.

Table 5: Grand Slam Lottery: Immediate Event-Level Effects

Outcome	<i>t</i> -test	Event FE	
	LL Mean	$\hat{\beta}$	(SE)
<i>ATP</i>			
Main-draw match win	0.34***	0.32***	(0.06)
Main-draw matches played	1.45***	1.48***	(0.08)
Ranking points at event	28.14***	29.66***	(3.79)
<i>N</i>	132 / 116	248	
<i>WTA</i>			
Main-draw match win	0.39***	0.38***	(0.06)
Main-draw matches played	1.43***	1.41***	(0.07)
Ranking points at event	25.28***	24.86***	(2.71)
<i>N</i>	54 / 78	132	
<i>Pooled</i>			
Main-draw match win	0.35***	0.36***	(0.04)
Main-draw matches played	1.45***	1.47***	(0.05)
Ranking points at event	27.31***	28.46***	(2.43)
<i>N</i>	186 / 194	380	

Notes. Full Grand Slam sample: top-four ranked final-round qualifying losers at events with at least one LL slot, 2006–2024. ATP: $N = 248$; WTA: $N = 132$. Event FE columns report OLS with Grand Slam $\times$ Year fixed effects. Control outcomes are mechanically zero because unselected players do not enter the main draw. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.2 Post-Episode Dynamics

Tables 6 and 7 report estimates from a stacked specification with horizon-specific treatment effects estimated jointly. The stacked model pools all five horizons into a single regression with horizon-specific LL coefficients and horizon fixed effects, improving efficiency by borrowing information across horizons while allowing the treatment effect to vary freely at each horizon. I present ATP and WTA results separately.

ATP results. The ATP sample comprises 248 player-episodes contributing approximately 1,200 stacked observations across five horizons. Ranking points increase by 31.9 at 4 weeks ($p = 0.028$), 38.5 at 12 weeks ($p = 0.007$), and 49.2 at 26 weeks ($p = 0.054$). The point gain fades to 4.3 at 52 weeks ($p = 0.919$), consistent with the rolling 52-week window replacing the initial point gain with subsequent results. Main draw entries increase by 0.6 at 12 weeks ($p = 0.046$) and 1.1 at 26 weeks ($p = 0.006$), and matches at 250+ events increase by 0.8 at 12 weeks ($p = 0.110$) and 1.8 at 26 weeks ($p = 0.010$). Elo rating changes are null at every horizon. The pattern confirms that the ranking point boost from the LL event is real but temporary: it builds through 26 weeks as the player enters additional tournaments, then dissipates as the initial points age out of the rolling window.

WTA results. The WTA sample comprises 132 player-episodes contributing approximately 640 stacked observations. Ranking points increase sharply at 4 weeks (64.4, $p = 0.003$) and 8 weeks (78.7, $p < 0.001$), but the effect dissipates faster than on the ATP tour: the 12-week estimate is 32.3 ($p = 0.135$) and the 26-week estimate is 6.3 ($p = 0.861$), consistent with the WTA’s denser tournament calendar accelerating the point cycle. Elo effects are positive at 4 weeks (10.5, $p = 0.070$) and 26 weeks (17.3, $p = 0.072$) but do not reach conventional significance at most horizons. The WTA results should be interpreted with caution given the smaller sample and borderline balance ($p = 0.333$).

Figures 1 and 2 plot mean cumulative ranking points earned by LL-selected and unselected players from 12 weeks before to 52 weeks after each Grand Slam. The pre-event trajectories are parallel on both tours, confirming the absence of differential pre-trends. After the event, ATP LL recipients accumulate approximately 32 additional ranking points by 4 weeks and 49 by 26 weeks, before the advantage narrows at 52 weeks as the rolling window replaces the initial gains. WTA LL recipients show a sharper initial divergence (64 points at 4 weeks, 79 at 8 weeks) but control players catch up more rapidly, consistent with the denser WTA tournament calendar.

Table 6: ATP Grand Slam: Post-Episode Dynamic Effects (Stacked Model)

	4w	8w	12w	26w	52w
Ranking Pts Δ	31.9** (14.5)	31.3** (14.2)	38.5*** (14.3)	49.2* (25.6)	4.3 (42.1)
Elo Δ	-1.6 (5.1)	0.2 (5.9)	-3.9 (6.1)	2.8 (8.8)	-14.5 (12.8)
Main Draws	0.2 (0.3)	0.5* (0.3)	0.6** (0.3)	1.1*** (0.4)	1.3 (0.9)
Matches 250+	0.2 (0.6)	0.6 (0.5)	0.8* (0.5)	1.8*** (0.7)	2.2 (1.8)
Win %	-0.01 (0.03)	-0.01 (0.02)	-0.03 (0.02)	-0.02 (0.02)	-0.02 (0.02)

Notes. Stacked specification: 248 player-episodes contributing 1,200–1,240 stacked observations across five horizons (4, 8, 12, 26, 52 weeks). Horizon-specific LL treatment effects estimated jointly with horizon fixed effects, Grand Slam $\times$ Year fixed effects, and pre-treatment controls (ranking points linear and quadratic, Elo / 100 linear and quadratic, prior GS LL spots won, prior GS LL candidacies not won, prior non-GS LL spots won, prior non-GS LL candidacies not won, and age). Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 7: WTA Grand Slam: Post-Episode Dynamic Effects (Stacked Model)

	4w	8w	12w	26w	52w
Ranking Pts Δ	64.4*** (21.4)	78.7*** (22.2)	32.3 (21.6)	6.3 (36.0)	6.3 (66.6)
Elo Δ	10.5* (5.8)	7.4 (6.7)	6.5 (8.0)	17.3* (9.6)	13.4 (16.1)
Main Draws	0.1 (0.3)	0.5 (0.3)	0.5 (0.3)	0.8* (0.5)	1.2 (1.1)
Matches 250+	-0.2 (0.5)	0.9* (0.5)	0.9* (0.5)	1.6** (0.8)	1.7 (1.9)
Win %	0.05 (0.05)	0.01 (0.03)	0.00 (0.03)	-0.00 (0.03)	0.01 (0.02)

Notes. Stacked specification: 132 player-episodes contributing 640–660 stacked observations across five horizons (4, 8, 12, 26, 52 weeks). Horizon-specific LL treatment effects estimated jointly with horizon fixed effects, Grand Slam $\times$ Year fixed effects, and pre-treatment controls. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

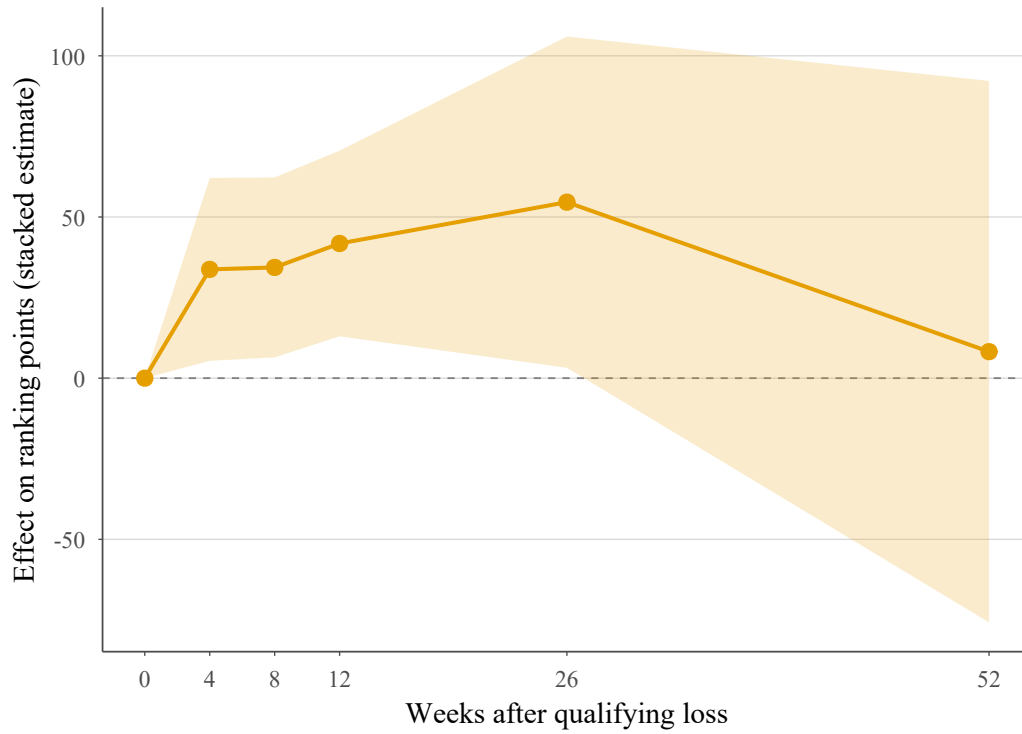


Figure 1: ATP Event Study: Ranking Trajectories for Selected and Unselected Players

Notes. Each point represents the estimated treatment effect at the indicated horizon. Vertical dashed line marks the event date. Shaded regions are 95 percent confidence intervals. Full ATP Grand Slam sample ($N = 248$).

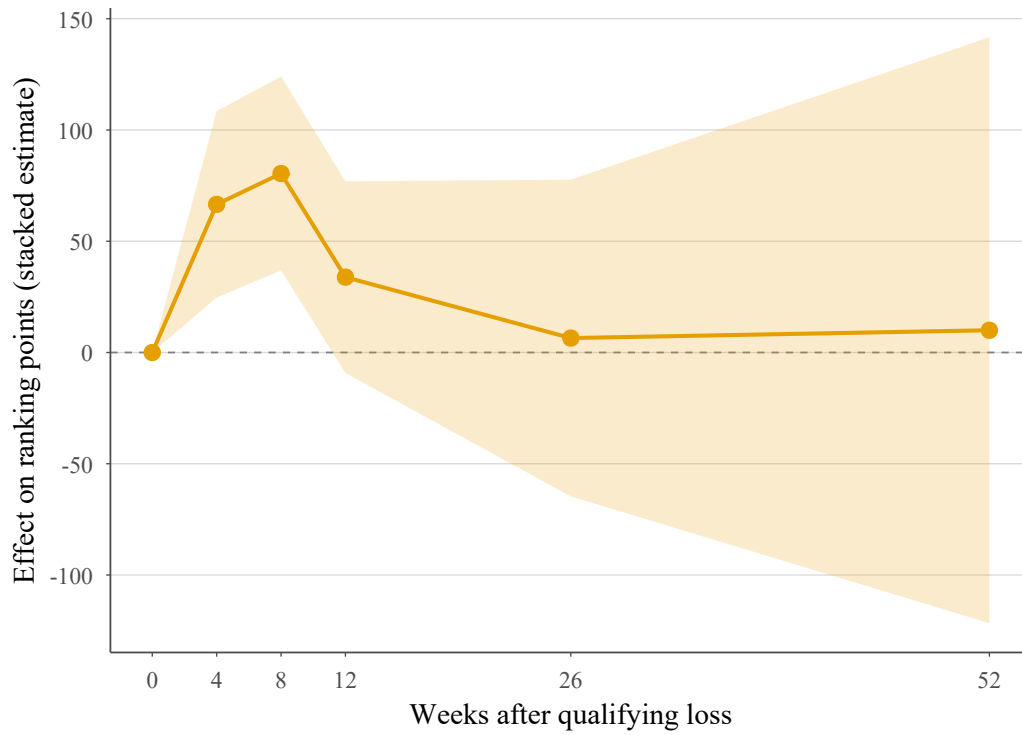


Figure 2: WTA Event Study: Ranking Trajectories for Selected and Unselected Players

Notes. Each point represents the estimated treatment effect at the indicated horizon. Full WTA Grand Slam sample ($N = 132$).

5.3 Secondary Outcomes: Elo and Predicted Win Probability

Elo ratings are unchanged at every horizon on the ATP tour. The stacked Elo coefficient ranges from -1.6 at 4 weeks ($p = 0.754$) to -14.5 at 52 weeks ($p = 0.257$)—none close to statistical significance. On the WTA tour, the Elo coefficient is marginally significant at 4 weeks (10.5 , $p = 0.070$) and 26 weeks (17.3 , $p = 0.072$) but does not reach conventional significance at other horizons. The Elo null across the medium- and long-run horizons is the cleanest test of ability: Elo adjusts for opponent quality, so if LL entry improved playing skill through experience against stronger opponents, Elo ratings would rise. They do not. The null Elo result, combined with the positive access effects, establishes that the lottery shock operates through tournament access rather than through improved match performance.

The predicted win probability $P(i \rightarrow j)$, computed from the Elo-based model, is also unchanged. This is a direct implication of the Elo null: if the focal player’s Elo does not move, their expected probability of beating a given opponent is unchanged. The two results together confirm that the access gains documented in Section 5.2 do not translate into improved quality-adjusted performance.

5.4 Heterogeneous Effects

Tables 8 and 9 report heterogeneity in the treatment effect, estimated separately by subgroup within each tour.

By age. Younger players (under 24) show larger point estimates than older players, though the differential is imprecisely estimated. The age gradient is consistent with younger players having more career runway over which to convert a ranking boost into sustained access.

By pre-episode ranking category. Effects are imprecisely estimated across ranking subgroups, with no significant differences between high- and low-ranked players within the LL candidate pool.

By prior LL experience. Players without a prior LL slot show larger and more persistent effects than those with prior experience. First-time LL recipients gain 73.8 ranking points at 12 weeks ($p < 0.001$) and 91.3 points at 26 weeks ($p = 0.004$), while the differential for repeat LL players is large and negative (-87.9 points at 26 weeks, $p = 0.015$). This suggests diminishing returns to repeated LL opportunities: the first break matters most, consistent with the labor market analog of a single career-altering opportunity.

By treatment dose. The dose-response analysis is the most informative dimension of heterogeneity. Table 10 reports horizon-specific estimates of the LL base effect (zero main draw wins) and the $LL \times$ matches won interaction, where matches won enters continuously. The base LL effect—receiving a slot without winning any main draw matches—is positive but imprecisely estimated: 37.3 points at 26 weeks ($p = 0.154$). The continuous matches-won interaction is 21.6 additional points per match won at 26 weeks ($p = 0.327$), though it is imprecisely estimated. At shorter horizons, the interaction is 27.7 at 4 weeks ($p = 0.111$) and 25.3 at 12 weeks ($p = 0.166$). The pattern indicates that while the initial LL slot provides some ranking point benefit, the treatment effect is amplified when the player wins matches in the main draw.

Performance probability interaction. I also test whether the magnitude of the career effects varies with the probability of the observed LL performance. For each LL recipient, I compute the sequential probability of their observed tournament trajectory—the product of match-level win/loss probabilities from the match-winning logit—and transform it to the performance dose $-\text{logit}(\pi_{ie}) = \log((1 - \pi_{ie})/\pi_{ie})$. A higher dose indicates a more surprising performance. Figure 3 shows the distribution of π_{ie} for LL recipients at Grand Slam and non-Grand-Slam events. Interacting the LL indicator with this dose in the post-episode stacked specification tests whether more surprising performances generate larger subsequent career benefits. Tables 12 and 13 report these results.

Table 8: ATP Grand Slam: Heterogeneous Effects

	4w		8w		12w		26w		
	β_h	$\beta_h \times X_k$	β_h	$\beta_h \times X_k$	β_h	$\beta_h \times X_k$	β_h	$\beta_h \times X_k$	β_h
<i>Interaction: Above-Median Rank</i>									
Ranking Points Δ	35.89** (16.79)	-12.83 (20.00)	30.49* (17.60)	-4.63 (22.33)	35.80* (18.94)	4.85 (25.63)	25.04 (29.85)	47.76 (38.47)	11.11 (42.11)
Elo Δ	-2.91 (6.89)	1.56 (7.28)	-5.54 (7.76)	9.51 (8.49)	-9.22 (8.42)	9.17 (10.30)	-5.04 (10.42)	19.54 (15.83)	0.14 (14.14)
Main Draws	0.27 (0.31)	-0.01 (0.35)	0.45 (0.32)	0.15 (0.36)	0.57* (0.33)	0.23 (0.39)	0.44 (0.49)	1.35** (0.63)	0.14 (1.14)
Matches (250+)	0.38 (0.64)	-0.13 (0.72)	0.61 (0.61)	0.19 (0.72)	0.69 (0.62)	0.42 (0.77)	0.46 (0.84)	2.46** (1.17)	1.21 (2.21)
Win %	0.01 (0.05)	-0.04 (0.06)	-0.00 (0.03)	-0.02 (0.04)	-0.01 (0.03)	-0.03 (0.03)	-0.01 (0.03)	-0.01 (0.03)	0.01 (0.01)
<i>Interaction: Above-Median Age</i>									
Ranking Points Δ	40.80** (16.10)	-24.05 (22.45)	34.31** (17.31)	-14.72 (25.64)	45.63** (20.91)	-16.92 (30.09)	60.40* (33.22)	-26.38 (41.74)	48.11 (57.11)
Elo Δ	0.89 (5.96)	-7.40 (8.31)	4.25 (8.31)	-11.94 (9.69)	0.36 (8.74)	-11.06 (11.60)	9.99 (12.52)	-12.52 (16.88)	14.14 (17.14)
Main Draws	0.04 (0.27)	0.60 (0.41)	0.19 (0.27)	0.82* (0.44)	0.17 (0.30)	1.15** (0.47)	0.66 (0.46)	1.01 (0.67)	0.14 (1.14)
Matches (250+)	0.25 (0.57)	0.28 (0.80)	0.63 (0.56)	0.26 (0.84)	0.59 (0.61)	0.71 (0.90)	1.63* (0.94)	0.20 (1.38)	1.21 (2.21)
Win %	0.02 (0.04)	-0.08 (0.06)	-0.01 (0.03)	-0.01 (0.04)	-0.02 (0.03)	-0.02 (0.04)	-0.01 (0.02)	-0.02 (0.03)	0.01 (0.01)
<i>Interaction: Had Prior LL</i>									
Ranking Points Δ	52.27*** (16.27)	-47.63** (20.40)	57.60*** (18.01)	-60.29*** (21.74)	73.77*** (19.44)	-72.03*** (24.22)	91.28*** (31.95)	-87.85** (36.18)	-2.11 (52.11)
Elo Δ	-0.49 (6.25)	-3.57 (7.27)	2.80 (7.66)	-7.83 (9.15)	-2.84 (8.09)	-3.80 (11.57)	10.59 (11.80)	-12.25 (17.25)	-16.16 (16.16)
Main Draws	0.40 (0.31)	-0.19 (0.40)	0.63** (0.31)	-0.13 (0.42)	0.66** (0.32)	0.13 (0.39)	1.04** (0.46)	0.22 (0.56)	0.14 (1.14)
Matches (250+)	0.87 (0.59)	-1.02 (0.79)	1.31** (0.60)	-1.13 (0.81)	1.34** (0.60)	-0.81 (0.75)	1.89** (0.88)	-0.33 (1.12)	1.21 (2.21)
Win %	0.03 (0.04)	-0.11* (0.06)	-0.00 (0.03)	-0.02 (0.04)	-0.01 (0.02)	-0.05 (0.04)	-0.01 (0.02)	-0.02 (0.03)	-0.01 (0.01)
Z^{pre} controls					Yes (horizon-interacted)				

Notes. Stacked specification with horizon-specific LL treatment effects estimated by subgroup. Full ATP Grand Slam sample. Low/High rank points split at the tour-specific median. Younger/Older split at age 24. Standard errors clustered at player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 9: WTA Grand Slam: Heterogeneous Effects

	4w		8w		12w		26w		
	β_h	$\beta_h \times X_k$	β_h	$\beta_h \times X_k$	β_h	$\beta_h \times X_k$	β_h	$\beta_h \times X_k$	β_h
<i>Interaction: Above-Median Rank</i>									
Ranking Points Δ	75.77*** (23.56)	-48.01 (37.25)	88.15*** (25.86)	-48.35 (42.60)	53.65** (26.18)	-68.30 (42.75)	57.55 (39.69)	-93.53 (69.72)	132.5 (68.1)
Elo Δ	11.03 (8.94)	-6.81 (12.39)	13.02 (10.19)	-14.96 (13.73)	7.84 (11.22)	-10.57 (15.42)	38.16*** (13.60)	-35.96* (18.27)	40.03 (21.6)
Main Draws	0.49 (0.46)	-0.60 (0.68)	0.39 (0.48)	0.02 (0.66)	0.23 (0.52)	0.24 (0.70)	0.79 (0.72)	-0.01 (0.98)	2.80 (1.71)
Matches (250+)	0.68 (0.67)	-0.49 (1.13)	1.08* (0.55)	-0.19 (0.98)	0.59 (0.55)	0.42 (0.94)	1.31 (0.88)	0.46 (1.60)	3.21 (2.34)
Win %	0.00 (0.07)	0.10 (0.09)	0.04 (0.05)	-0.03 (0.07)	0.03 (0.05)	-0.04 (0.07)	0.04 (0.03)	-0.10* (0.05)	0.04 (0.03)
<i>Interaction: Above-Median Age</i>									
Ranking Points Δ	17.11 (23.01)	68.44* (39.31)	23.88 (26.33)	78.75* (41.35)	-28.07 (31.56)	93.25** (44.57)	1.90 (53.01)	21.91 (58.69)	83.9 (76.3)
Elo Δ	4.85 (7.68)	5.56 (11.47)	1.26 (10.89)	7.22 (13.99)	-5.77 (12.89)	15.41 (15.40)	33.65** (15.35)	-21.80 (18.28)	28.6 (24.6)
Main Draws	0.23 (0.43)	-0.10 (0.62)	0.52 (0.46)	-0.19 (0.64)	0.34 (0.54)	0.07 (0.68)	0.30 (0.70)	0.92 (0.95)	1.21 (1.64)
Matches (250+)	0.35 (0.69)	0.20 (1.01)	0.94 (0.66)	0.18 (0.91)	0.43 (0.75)	0.83 (0.93)	0.96 (1.19)	1.21 (1.40)	-0.13 (2.42)
Win %	0.02 (0.06)	0.08 (0.10)	-0.02 (0.05)	0.07 (0.07)	-0.04 (0.04)	0.08 (0.06)	-0.02 (0.04)	0.03 (0.05)	0.00 (0.03)
<i>Interaction: Had Prior LL</i>									
Ranking Points Δ	40.57* (21.64)	30.79 (49.18)	58.73** (22.89)	14.23 (52.82)	25.40 (24.30)	-17.27 (53.01)	31.12 (44.82)	-57.66 (65.20)	81.7 (72.1)
Elo Δ	10.81 (6.83)	-6.65 (10.81)	12.29 (8.59)	-18.61 (11.59)	5.37 (9.89)	-6.23 (14.15)	23.41* (12.79)	-5.29 (15.06)	33.37 (19.0)
Main Draws	0.11 (0.39)	0.15 (0.68)	0.34 (0.41)	0.21 (0.64)	0.31 (0.45)	0.20 (0.63)	0.69 (0.61)	0.30 (0.84)	1.48 (1.51)
Matches (250+)	0.32 (0.61)	0.30 (1.14)	1.11** (0.51)	-0.31 (1.07)	0.76 (0.55)	0.22 (0.98)	1.60* (0.93)	-0.09 (1.42)	1.42 (2.28)
Win %	0.07 (0.05)	-0.01 (0.12)	0.06 (0.04)	-0.12** (0.06)	0.04 (0.03)	-0.10 (0.06)	0.02 (0.03)	-0.08 (0.05)	0.03 (0.03)
Z^{pre} controls					Yes (horizon-interacted)				

Notes. Each row reports a separate regression. Full WTA Grand Slam sample, 2006–2024. Standard errors clustered at tournament level.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 10: Treatment Dose: LL Base Effect and LL $\times$ Matches Won Interaction (Stacked)

	4w	8w	12w	26w	52w
<i>Panel A: Matches won dose</i>					
Ranking Points Δ (base)	17.53 (14.13)	16.34 (14.92)	25.92 (15.89)	37.32 (26.15)	-19.30 (40.64)
$\times$ Matches Won	27.66 (17.34)	24.53 (17.89)	25.29 (18.26)	21.61 (22.05)	42.63 (42.75)
Elo Δ (base)	-1.89 (5.61)	0.02 (6.50)	-3.70 (7.05)	5.62 (9.46)	-13.97 (13.25)
$\times$ Matches Won	-0.18 (5.90)	-3.02 (6.03)	-2.65 (6.99)	-3.78 (11.69)	3.19 (12.72)
Main Draws (base)	0.46* (0.28)	0.73*** (0.27)	0.87*** (0.27)	1.20*** (0.39)	0.82 (0.93)
$\times$ Matches Won	-0.44 (0.29)	-0.45* (0.27)	-0.38 (0.28)	-0.07 (0.38)	0.42 (0.97)
Matches (250+) (base)	0.53 (0.53)	0.90* (0.51)	1.01** (0.51)	1.76** (0.72)	1.09 (1.68)
$\times$ Matches Won	-0.50 (0.62)	-0.47 (0.54)	-0.19 (0.56)	-0.10 (0.77)	1.12 (1.83)
Win % (base)	-0.03 (0.03)	-0.02 (0.02)	-0.04 (0.02)	-0.02 (0.02)	-0.02 (0.02)
$\times$ Matches Won	0.03 (0.06)	0.03 (0.03)	0.02 (0.02)	0.00 (0.02)	0.00 (0.02)
<i>Panel B: Performance probability dose ($-\text{logit}(\pi)$)</i>					
Ranking Points Δ (base)	35.22** (14.28)	35.28** (14.58)	47.04*** (15.12)	58.69** (26.34)	11.40 (42.26)
$\times$ Dose	2.92 (8.23)	7.30 (7.96)	13.61 (8.89)	22.43 (14.70)	35.50** (15.29)
Elo Δ (base)	-0.41 (5.31)	1.22 (6.40)	-1.23 (6.57)	6.70 (10.04)	-8.34 (12.68)
$\times$ Dose	0.99 (2.52)	2.42 (2.51)	5.78* (3.02)	3.73 (6.10)	14.21** (5.88)
Main Draws (base)	0.22 (0.28)	0.50* (0.27)	0.67** (0.27)	1.18*** (0.38)	0.99 (0.89)
$\times$ Dose	-0.22 (0.15)	-0.15 (0.16)	-0.10 (0.16)	0.16 (0.18)	0.32 (0.38)
Matches (250+) (base)	0.38 (0.53)	0.82* (0.49)	1.04** (0.50)	1.90*** (0.72)	1.68 (1.70)
$\times$ Dose	-0.17 (0.23)	0.02 (0.25)	0.11 (0.26)	0.37 (0.32)	0.72 (0.65)
Win % (base)	-0.01 (0.03)	-0.00 (0.02)	-0.02 (0.02)	-0.01 (0.02)	-0.00 (0.02)
$\times$ Dose	0.01 (0.02)	0.01 (0.01)	0.02* (0.01)	0.01 (0.01)	0.02** (0.01)
Z^{pre} controls	30 Yes (horizon-interacted)				

Notes. Stacked specification with horizon-specific coefficients on the LL indicator and the LL $\times$ matches won (continuous) interaction. Full ATP Grand Slam sample, 2006–2024. The LL indicator is equal to 1 if the player is a left-handed player.

Table 11: Treatment Dose: Non-Grand-Slam (Stacked)

	4w	8w	12w	26w	52w
<i>Panel A: Matches won dose</i>					
Ranking Points Δ (base)	-0.46 (6.55)	-7.34 (7.46)	-14.22* (8.29)	-4.22 (11.02)	-46.56* (23.85)
× Matches Won	26.66*** (4.83)	36.62*** (7.30)	38.90*** (7.74)	60.15*** (12.45)	87.91*** (21.09)
Elo Δ (base)	0.17 (3.81)	2.49 (3.88)	1.77 (4.18)	7.89 (8.20)	6.31 (13.91)
× Matches Won	1.36 (1.46)	1.30 (2.08)	1.28 (2.49)	0.02 (3.22)	2.84 (3.92)
Main Draws (base)	-1.22*** (0.17)	-0.72*** (0.15)	-0.19 (0.14)	1.57*** (0.22)	4.10*** (0.58)
× Matches Won	-0.50*** (0.11)	-0.42*** (0.11)	-0.33*** (0.11)	0.15 (0.17)	1.34*** (0.35)
Matches (250+) (base)	-2.21*** (0.32)	-1.36*** (0.28)	-0.51* (0.27)	2.58*** (0.40)	6.70*** (1.02)
× Matches Won	-1.06*** (0.23)	-0.88*** (0.21)	-0.63*** (0.22)	0.34 (0.34)	2.81*** (0.75)
Win % (base)	-0.04* (0.02)	-0.03** (0.01)	-0.02 (0.02)	-0.01 (0.02)	-0.00 (0.02)
× Matches Won	-0.01 (0.01)	-0.01 (0.01)	0.01 (0.01)	-0.01 (0.01)	-0.01 (0.01)
<i>Panel B: Performance probability dose ($-\text{logit}(\pi)$)</i>					
Ranking Points Δ (base)	12.45** (6.19)	11.11 (6.83)	5.93 (7.61)	25.96** (10.47)	2.83 (22.66)
× Dose	13.99*** (3.25)	15.28*** (3.79)	17.54*** (4.41)	32.69*** (7.69)	46.43*** (14.24)
Elo Δ (base)	1.12 (3.72)	3.53 (3.90)	2.96 (4.17)	8.07 (7.92)	8.92 (13.31)
× Dose	0.38 (1.10)	-1.50 (1.37)	-2.94** (1.46)	0.69 (2.02)	0.91 (2.84)
Main Draws (base)	-1.47*** (0.16)	-0.94*** (0.14)	-0.35*** (0.12)	1.67*** (0.21)	4.91*** (0.55)
× Dose	-0.29*** (0.08)	-0.22*** (0.08)	-0.19** (0.08)	0.04 (0.11)	0.60** (0.26)
Matches (250+) (base)	-2.76*** (0.29)	-1.81*** (0.26)	-0.82*** (0.24)	2.79*** (0.39)	8.37*** (0.98)
× Dose	-0.64*** (0.15)	-0.56*** (0.14)	-0.42*** (0.15)	0.13 (0.22)	1.32** (0.55)
Win % (base)	-0.05** (0.02)	-0.03** (0.01)	-0.01 (0.02)	-0.01 (0.02)	-0.01 (0.02)
× Dose	0.01 (0.01)	-0.00 (0.01)	-0.01 (0.01)	-0.00 (0.00)	-0.00 (0.00)
Z^{pre} controls	31 Yes (main effects)				

Notes. Same specification as the GS dose table with control function correction. Matches won is NOT centered. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 12: Performance Probability Dose: Grand Slam (Stacked)

	4w	8w	12w	26w	52w
<i>Panel A: Matches won dose</i>					
Ranking Points Δ (base)	32.73 (20.35)	32.07 (22.70)	-10.96 (26.39)	-41.78 (43.83)	-15.27 (68.11)
× Matches Won	33.46 (36.78)	83.80** (40.37)	76.60* (40.77)	144.52** (55.32)	221.02** (106.99)
Elo Δ (base)	1.70 (6.34)	-3.01 (7.47)	-6.40 (9.36)	11.66 (11.06)	10.92 (19.56)
× Matches Won	13.94 (10.81)	22.52* (12.40)	27.06* (13.93)	23.61 (15.03)	24.79 (23.36)
Main Draws (base)	0.34 (0.34)	0.56 (0.37)	0.53 (0.40)	0.64 (0.55)	0.04 (1.37)
× Matches Won	-0.63 (0.58)	-0.53 (0.58)	-0.57 (0.62)	0.41 (0.92)	3.20* (1.86)
Matches (250+) (base)	0.66 (0.52)	1.00* (0.53)	0.84 (0.55)	1.10 (0.90)	-1.57 (2.00)
× Matches Won	-1.06 (1.02)	-0.26 (0.96)	-0.30 (1.00)	1.24 (1.32)	7.75** (3.11)
Win % (base)	0.04 (0.06)	-0.01 (0.04)	-0.02 (0.04)	-0.03 (0.03)	0.01 (0.03)
× Matches Won	0.06 (0.09)	0.06 (0.06)	0.05 (0.05)	0.05 (0.04)	-0.00 (0.04)
<i>Panel B: Performance probability dose ($-\text{logit}(\pi)$)</i>					
Ranking Points Δ (base)	48.79*** (18.54)	64.71*** (20.77)	17.73 (23.31)	19.00 (38.57)	84.37 (62.34)
× Dose	-2.47 (13.59)	5.86 (16.53)	-1.92 (16.94)	24.84 (22.74)	47.63 (43.90)
Elo Δ (base)	6.43 (5.53)	5.65 (7.30)	0.62 (9.07)	23.49** (9.89)	21.71 (14.89)
× Dose	-2.88 (3.87)	2.10 (4.61)	-3.20 (5.12)	8.03 (5.88)	2.81 (10.12)
Main Draws (base)	0.08 (0.31)	0.30 (0.33)	0.24 (0.34)	0.75 (0.52)	1.19 (1.18)
× Dose	-0.26 (0.21)	-0.35* (0.20)	-0.39* (0.21)	-0.11 (0.30)	0.39 (0.76)
Matches (250+) (base)	0.26 (0.50)	0.87* (0.48)	0.74 (0.49)	1.69** (0.83)	1.80 (1.83)
× Dose	-0.41 (0.39)	-0.35 (0.33)	-0.25 (0.36)	0.30 (0.50)	2.41** (1.21)
Win % (base)	0.05 (0.06)	0.01 (0.04)	-0.01 (0.04)	-0.01 (0.03)	0.00 (0.02)
× Dose	-0.03 (0.04)	-0.02 (0.02)	-0.03 (0.02)	-0.01 (0.01)	-0.02 (0.01)
Z^{pre} controls	32 Yes (horizon-interacted)				

Notes. Dose = $-\text{logit}(\pi_{ie})$ where π_{ie} is the probability of observing the LL awardee's main-draw performance. Higher dose = more surprising performance. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 13: Performance Probability Dose: Non-Grand-Slam (Stacked)

	4w	8w	12w	26w	52w
<i>Panel A: Matches won dose</i>					
Ranking Points Δ (base)	12.91 (9.93)	4.34 (10.69)	2.78 (11.61)	-12.83 (17.59)	-15.81 (34.82)
× Matches Won	28.42*** (5.77)	33.67*** (6.41)	38.60*** (7.50)	34.01*** (12.94)	46.61 (30.43)
Elo Δ (base)	4.40* (2.65)	4.08 (2.88)	7.09** (3.05)	3.34 (4.83)	2.86 (6.68)
× Matches Won	4.04** (1.70)	2.45 (2.07)	3.39 (2.35)	0.14 (3.84)	-0.43 (6.52)
Main Draws (base)	-0.89*** (0.14)	-0.46*** (0.13)	-0.11 (0.12)	1.04*** (0.18)	2.40*** (0.46)
× Matches Won	-0.06 (0.10)	-0.03 (0.09)	0.06 (0.10)	0.34** (0.17)	1.58*** (0.46)
Matches (250+) (base)	-1.73*** (0.29)	-1.09*** (0.26)	-0.55** (0.25)	1.60*** (0.38)	4.66*** (0.91)
× Matches Won	-0.12 (0.20)	0.03 (0.18)	0.27 (0.20)	0.50 (0.33)	2.29*** (0.85)
Win % (base)	0.02 (0.02)	0.02 (0.01)	0.01 (0.01)	-0.00 (0.01)	0.00 (0.01)
× Matches Won	-0.00 (0.01)	-0.02 (0.01)	-0.01 (0.01)	-0.00 (0.01)	-0.01 (0.01)
<i>Panel B: Performance probability dose ($-\logit(\pi)$)</i>					
Ranking Points Δ (base)	28.46*** (9.50)	23.10** (10.20)	23.60** (11.30)	4.46 (15.62)	6.48 (31.01)
× Dose	13.64*** (3.77)	16.80*** (4.49)	19.39*** (5.01)	16.94* (9.38)	16.70 (17.97)
Elo Δ (base)	6.56*** (2.54)	5.34* (2.76)	8.75*** (2.93)	2.98 (4.37)	1.32 (5.94)
× Dose	2.52** (1.19)	1.37 (1.43)	2.72 (1.71)	0.23 (2.67)	-5.42 (3.75)
Main Draws (base)	-0.91*** (0.13)	-0.46*** (0.12)	-0.06 (0.11)	1.24*** (0.16)	3.22*** (0.41)
× Dose	-0.02 (0.07)	0.01 (0.06)	0.06 (0.07)	0.24** (0.11)	0.88*** (0.30)
Matches (250+) (base)	-1.78*** (0.26)	-1.06*** (0.25)	-0.39 (0.24)	1.89*** (0.36)	5.86*** (0.81)
× Dose	-0.07 (0.13)	0.05 (0.12)	0.20 (0.13)	0.32 (0.21)	1.17** (0.55)
Win % (base)	0.02 (0.02)	0.01 (0.01)	0.00 (0.01)	-0.00 (0.01)	-0.01 (0.01)
× Dose	0.01 (0.01)	0.00 (0.01)	0.00 (0.01)	0.00 (0.00)	0.00 (0.00)
Z^{pre} controls	33 Yes (main effects)				

Notes. Same specification as Table 12 with control function correction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

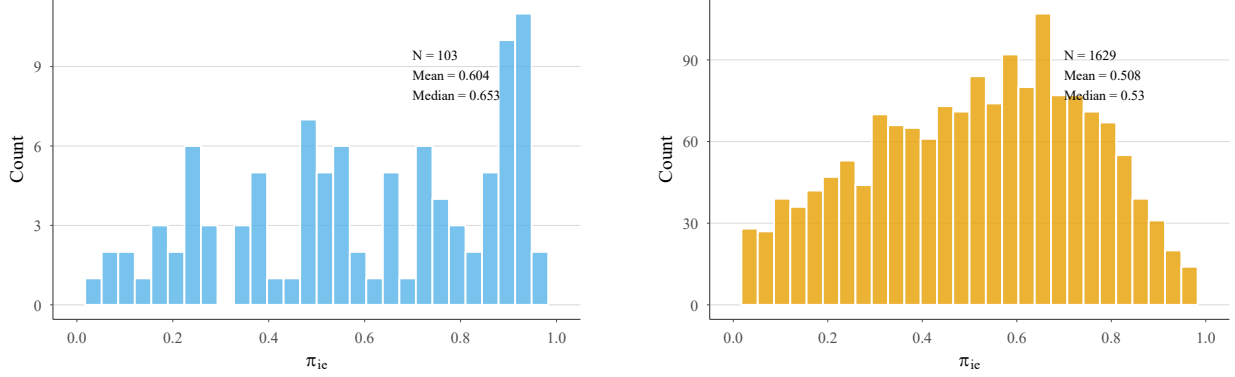


Figure 3: Distribution of Performance Probability π_{ie} for LL Recipients

Notes. Left panel: Grand Slam LL recipients. Right panel: non-Grand-Slam LL recipients. π_{ie} is the product of match-level outcome probabilities from the match-winning logit. Lower values indicate more surprising tournament performances.

5.5 Non-Grand-Slam Extension

The non-Grand-Slam control function estimates, which draw on the larger ATP sample ($N = 2,708$, with 874 treated) and the WTA sample ($N = 2,545$, with 755 treated), provide corroborating evidence. The generalized residual $\hat{v}_{ie}$ from the corrected selection probability P_i^{LL} —computed via exact enumeration of the full combinatorial structure of qualifying match pairings—enters all non-GS specifications to account for endogenous LL assignment.

ATP non-GS results. Ranking point gains are 12.2 at 4 weeks ($p = 0.065$), 11.0 at 8 weeks ($p = 0.127$), and 28.1 at 26 weeks ($p = 0.009$), fading to 6.5 at 52 weeks ($p = 0.776$). Tournament access effects emerge at longer horizons: main draw entries increase by 1.7 at 26 weeks ($p < 0.001$) and 5.0 at 52 weeks ($p < 0.001$), and matches at 250+ events increase by 2.8 at 26 weeks ($p < 0.001$) and 8.6 at 52 weeks ($p < 0.001$). Elo effects are null at all horizons. The endogeneity test (coefficient on $\hat{v}_{ie}$) is insignificant for ranking point outcomes (all $p > 0.10$), suggesting that OLS would be consistent for this outcome. However, $\hat{\rho}$ is significant for tournament access outcomes—main draws entered and matches at 250+ events—at short and long horizons ($p < 0.01$), confirming that the control function correction is necessary for count-based outcomes.

WTA non-GS results. The WTA non-Grand-Slam sample yields a ranking point gain of 27.4 at 4 weeks ($p = 0.004$) and 22.2 at 8 weeks ($p = 0.028$), with the effect persisting at 12 weeks (23.0, $p = 0.042$) before fading at 26 weeks (3.1, $p = 0.842$). The WTA access outcomes at 26 weeks are significant—main draw entries (+1.2, $p < 0.001$) and matches at 250+ events (+1.9, $p < 0.001$). The WTA non-GS Elo results differ from the Grand Slam pattern: Elo gains are significant at 4 weeks (6.6, $p = 0.008$), 8 weeks (5.5, $p = 0.050$), and 12 weeks (8.9, $p = 0.002$), fading at longer horizons. This short-horizon Elo effect on the WTA tour is consistent with the marginally significant WTA GS Elo result ($p = 0.070$ at 4 weeks), though the match-level tournament performance model (Section 6.3) finds no significant effect on win probability conditional on opponent strength on either tour. The WTA Elo gains at short horizons may reflect favorable opponent draws in additional tournaments rather than improved competitive skill per se.

The non-Grand-Slam CF estimates are broadly consistent with the Grand Slam lottery in direction and magnitude. The ATP CF finds ranking point gains of 28.1 points at 26 weeks ($p = 0.009$), compared to 49.2 points from the GS lottery ($p = 0.054$). Tournament access effects are directionally consistent: the CF finds 1.7 additional main draws at 26 weeks ($p < 0.001$), compared to 1.1 from the GS lottery ($p = 0.006$). The WTA non-GS results show stronger and more persistent short-horizon effects than the earlier estimates, with significant ranking point gains through 12 weeks and significant Elo gains through 12 weeks.

Full non-Grand-Slam tables are reported in the appendix: the ATP CF dynamics (Table 29), WTA CF dynamics (Table 32), heterogeneity (Tables 30 and 33), first-LL results (Tables 31 and 34), and the LL distribution (Table 24).

6 Mechanisms

The results establish that lucky loser entry causes gains in tournament access, ranking point accumulation, and no change in Elo ratings. This section synthesizes these findings through three channels: access, access cascade, and match-level performance conditional on opponent strength.

6.1 Access Channel

The most direct mechanism is that LL entry provides a main draw appearance that the player would not otherwise have received. This appearance generates ranking points proportional to the round reached, which in turn improve the player’s position in the rolling 52-week ranking. The improved ranking may cross direct-entry thresholds for subsequent tournaments—particularly at lower-tier events (ATP/WTB 250) where the entry threshold is more accessible. The evidence supports this channel: on the ATP tour, treated players enter 1.1 more main draws over 26 weeks ($p = 0.006$) and play 1.8 more matches at 250+ events ($p = 0.010$).

The Elo null establishes that the access channel operates through the ranking system, not through improved playing skill. Elo ratings adjust for opponent quality: a player who enters more tournaments but faces proportionally tougher opponents will not see their Elo rise unless their actual match performance improves. The null Elo result at all horizons, on both tours, confirms that the additional tournament access does not translate into better quality-adjusted match outcomes.

6.2 Access Cascade

The dose-response results trace how the access channel compounds through the ranking system. A player who receives the LL slot but loses in the first round earns minimal ranking points (10 at a Grand Slam). This is not enough to meaningfully change their position or

gain access to subsequent main draws. Their outcomes are indistinguishable from controls.

A player who wins one or more main draw matches earns substantially more points (45 for a second-round loss, 90 for a third-round loss at a Grand Slam). This shifts their position enough to cross direct-entry thresholds at lower-tier events, generating additional main draw entries over the next 26 weeks. Each subsequent entry provides another opportunity to earn points, creating a multiplier effect: LL entry $\rightarrow$ ranking points $\rightarrow$ more entries $\rightarrow$ more points $\rightarrow$ more entries.

The cascade is self-limiting. As the 52-week rolling window progresses, the points from the original LL event constitute a shrinking fraction of the player’s total. Unless downstream tournaments generate enough additional points to replace the original ones, the position converges back toward the level determined by sustained match performance. The dynamic results confirm this: ATP ranking point gains peak at 12–26 weeks and fade toward zero at 52 weeks.

6.3 Match-Level Performance Conditional on Opponent Strength

The reduced-form results show that LL assignment affects subsequent ranking points, tournament entry, and match volume. These outcomes combine multiple channels: opponent quality, tournament tier, and mechanical point accumulation. To isolate whether LL assignment affects *competitive performance itself*, I examine match-level win probabilities conditional on opponent strength and tournament context.

Conceptual framework. Consider player i who is in the LL candidate pool at event e and is subsequently observed in matches $m = 1, 2, \dots$ at tournaments $t > 0$. Each match outcome depends on the opponent faced and the competitive environment. The object of interest is whether, conditional on these factors, receiving an LL spot at event e changes the probability of winning future matches. Formally, the question is how LL assignment shifts $P(\text{win} \mid \text{opponent, context, pre-treatment ability})$.

Empirical specification. At Grand Slam events, LL assignment is by lottery, so D_{ie} is exogenous conditional on being in the candidate pool. I estimate a match-level logit:

$$p_{itm} = \Lambda(\beta' X_{ijm} + \gamma' Z_{ie}^{pre} + \delta \cdot D_{ie}) \quad (6)$$

where p_{itm} is the probability that player i wins match m against opponent j at tournament t and D_{ie} indicates LL receipt at event e . The match-level pairwise covariates X_{ijm} include ranking points difference / 1000 (linear and quadratic), Elo difference / 100 (linear and quadratic), surface Elo difference / 100 (linear and quadratic), Laplace-smoothed head-to-head win proportion, head-to-head encounter count, age difference, and clay and grass surface indicators. The pre-treatment covariates Z_{ie}^{pre} include ranking points / 1000 (linear and quadratic), Elo / 100 (linear and quadratic), surface Elo / 100 (linear and quadratic), and player age, all measured at the date of the qualifying loss at event e . Standard errors are clustered at the player level. No control function is needed because the lottery generates exogenous variation in D_{ie} .

For non-Grand-Slam events, where LL assignment is determined by ranking among qualifying losers, I use a control function approach. The generalized residual $\hat{v}_{ie}$, computed analytically from the selection probability P_i^{LL} (Section A.5), enters the outcome logit as an additional regressor:

$$p_{itm} = \Lambda(\beta' X_{ijm} + \gamma' Z_{ie}^{pre} + \delta \cdot D_{ie} + \rho \cdot \hat{v}_{ie}) \quad (7)$$

where X_{ijm} and Z_{ie}^{pre} are defined as in equation (6) and $\hat{v}_{ie}$ accounts for endogenous LL assignment. Those results are reported in Appendix B.4.

Identification. At Grand Slams, identification relies on the lottery: conditional on being in the top-four candidate pool, assignment is random. All focal-player covariates are measured at the date of the qualifying loss at event e (pre-treatment freezing), preventing

post-treatment variables from absorbing part of the causal pathway. Opponent characteristics are measured at the match date since they are unaffected by the focal player’s treatment.

Grand Slam results. The primary analysis includes all LL candidates—not only first-time candidates—to maximize statistical power from the lottery design. On the ATP tour ($N = 8,543$ matches from 130 unique players), the estimate is $\hat{\delta} = -0.123$ in log-odds (bootstrap SE = 0.079, $p = 0.119$). The point estimate is negative but not statistically significant, indicating no detectable effect of LL entry on subsequent match-level performance. On the WTA tour ($N = 5,391$ matches from 98 unique players), the estimate is $\hat{\delta} = +0.034$ (bootstrap SE = 0.112, $p = 0.760$), a small positive estimate far from statistical significance. Neither tour shows a significant match-level performance effect of LL entry, consistent with the interpretation that the career benefits documented in Section 5 operate through additional tournament access rather than improved competitive ability.

Tables 14 and 15 report the Grand Slam results for the ATP and WTA tours, respectively. Figure 4 traces the match-level treatment effect and derived tournament-level quantities across calendar horizons for the GS-ATP sample.

Dose-response analysis. Tables 16 and 17 report how the match-level treatment effect varies with the number of main draw matches won at the focal LL event. The dose variable is centered at the mean among treated players, so the baseline coefficient $\hat{\delta}$ captures the effect at the mean dose.

Horizon heterogeneity. Tables 18 and 19 allow the match-level treatment effect to vary by calendar horizon (4, 8, 12, 26, and 52 weeks after the focal event). On the ATP tour, the effect is null at every horizon. On the WTA tour, the effect is also null across horizons, with no statistically significant estimate at any time window.

Table 14: Grand Slam Tournament Performance: ATP

GS-ATP	
$\hat{\delta}$ (LL entry)	-0.1227 (0.0787)
Matches	8,543
Players	130

Notes. Logit estimates of the effect of LL entry on match win probability at subsequent tournaments. Grand Slam ATP sample: 8,543 matches from 130 unique players. Sample includes all matches (main draw, qualifying, and challenger events) within each episode's observation window. $\hat{\delta}$ is the coefficient on the LL indicator in log-odds units. Match-level covariates X_{ijm} include ranking points difference / 1000 (linear and quadratic), Elo difference / 100 (linear and quadratic), surface Elo difference / 100 (linear and quadratic), H2H win rate, H2H encounter count, age difference, and clay and grass surface indicators. Pre-treatment covariates Z_{ie}^{pre} include ranking points / 1000 (linear and quadratic), Elo / 100 (linear and quadratic), surface Elo / 100 (linear and quadratic), and player age. Bootstrap standard errors (200 replications) clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 15: Grand Slam Tournament Performance: WTA

GS-WTA	
$\hat{\delta}$ (LL entry)	0.0342 (0.1118)
Matches	5,391
Players	98

Notes. Logit estimates of the effect of LL entry on match win probability at subsequent tournaments. Grand Slam WTA sample: 5,391 matches from 98 unique players. $\hat{\delta}$ is the coefficient on the LL indicator in log-odds units. Match-level covariates X_{ijm} and pre-treatment covariates Z_{ie}^{pre} as in Table 14. Bootstrap standard errors (200 replications) clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 16: Grand Slam Dose-Response: ATP (Centered)

GS-ATP	
$\hat{\delta}$ (LL entry)	-0.1172** (0.0492)
$\hat{\delta} \times$ Matches won	0.0791 (0.0545)
Matches	8,543
Players	130

Notes. Logit estimates with LL treatment interacted with centered matches won at the focal event. Grand Slam ATP sample. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 17: Grand Slam Dose-Response: WTA (Centered)

GS-WTA	
$\hat{\delta}$ (LL entry)	0.0337 (0.0673)
$\hat{\delta} \times \text{Matches won}$	-0.0461 (0.1051)
Matches	5,391
Players	98

Notes. Logit estimates with LL treatment interacted with centered matches won at the focal event. Grand Slam WTA sample. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 18: Grand Slam Horizon Heterogeneity: ATP

	4w	8w	12w	26w	52w
δ_h (incremental)	0.0125 (0.1571)	0.0547 (0.1956)	-0.1376 (0.1577)	0.0652 (0.1057)	-0.1327 (0.0920)
Total effect	-0.1380 (0.1104)	-0.1505 (0.1403)	-0.2051 (0.1734)	-0.0676 (0.1098)	-0.1327 (0.0920)
Matches	8,543				

Notes. Logit estimates with horizon-specific LL treatment effects. Grand Slam ATP sample. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 19: Grand Slam Horizon Heterogeneity: WTA

	4w	8w	12w	26w	52w
δ_h (incremental)	0.3332 (0.2185)	-0.2591 (0.2851)	0.1636 (0.2521)	-0.1690 (0.1781)	0.0866 (0.1418)
Total effect	0.1553 (0.1669)	-0.1779 (0.2109)	0.0812 (0.2028)	-0.0824 (0.1461)	0.0866 (0.1418)
Matches	5,391				

Notes. Logit estimates with horizon-specific LL treatment effects. Grand Slam WTA sample. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

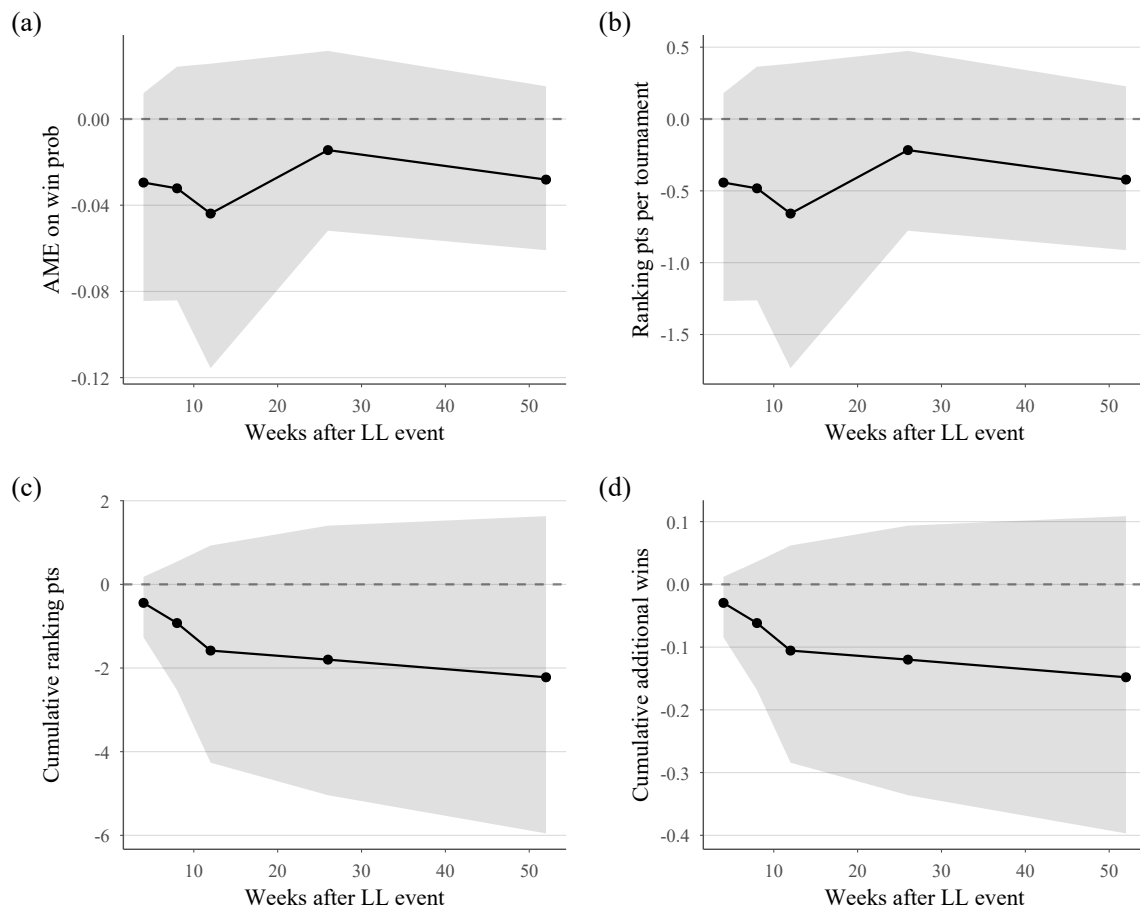


Figure 4: Match-Level Treatment Effects by Calendar Horizon: Grand Slam ATP

Notes. Four panels report the match-level LL treatment effect and derived tournament-level quantities at calendar horizons of 4, 8, 12, 26, and 52 weeks after the LL-granting event. Panel (a): average marginal effect on match win probability. Panel (b): expected additional ranking points per tournament. Panel (c): cumulative expected ranking points through each horizon. Panel (d): cumulative expected additional wins through each horizon. Grand Slam ATP sample ($N = 8,543$ matches, 130 players). Shaded bands are 95% confidence intervals from player-level block bootstrap (200 replications).

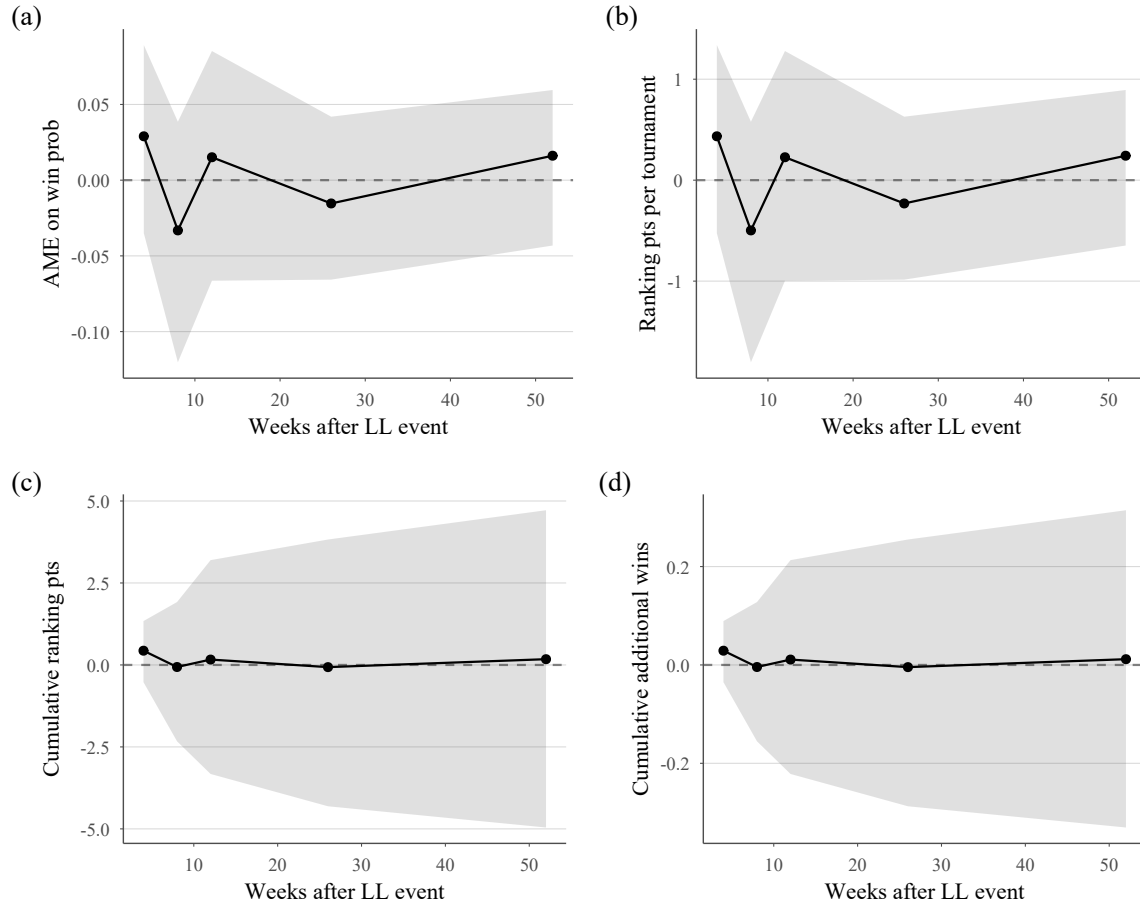


Figure 5: Match-Level Treatment Effects by Calendar Horizon: Grand Slam WTA
Notes. Same specification as Figure 4. Grand Slam WTA sample ($N = 5,391$ matches, 98 players). Shaded bands are 95% confidence intervals from player-level block bootstrap (200 replications).

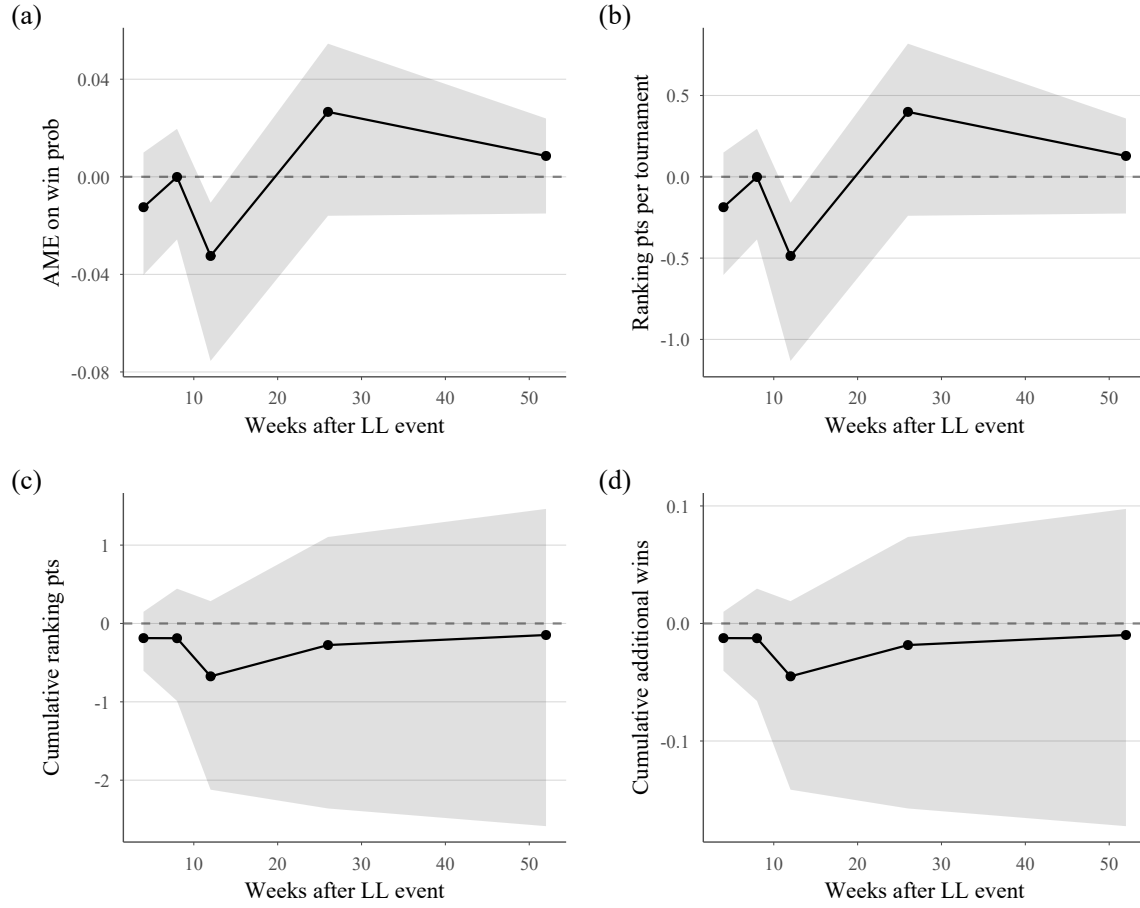


Figure 6: Match-Level Treatment Effects by Calendar Horizon: Non-Grand-Slam ATP

Notes. Same specification as Figure 4, with control function correction for endogenous LL assignment. Non-Grand-Slam ATP sample ($N = 113,189$ matches, 838 players). Shaded bands are 95% confidence intervals from player-level block bootstrap (200 replications).

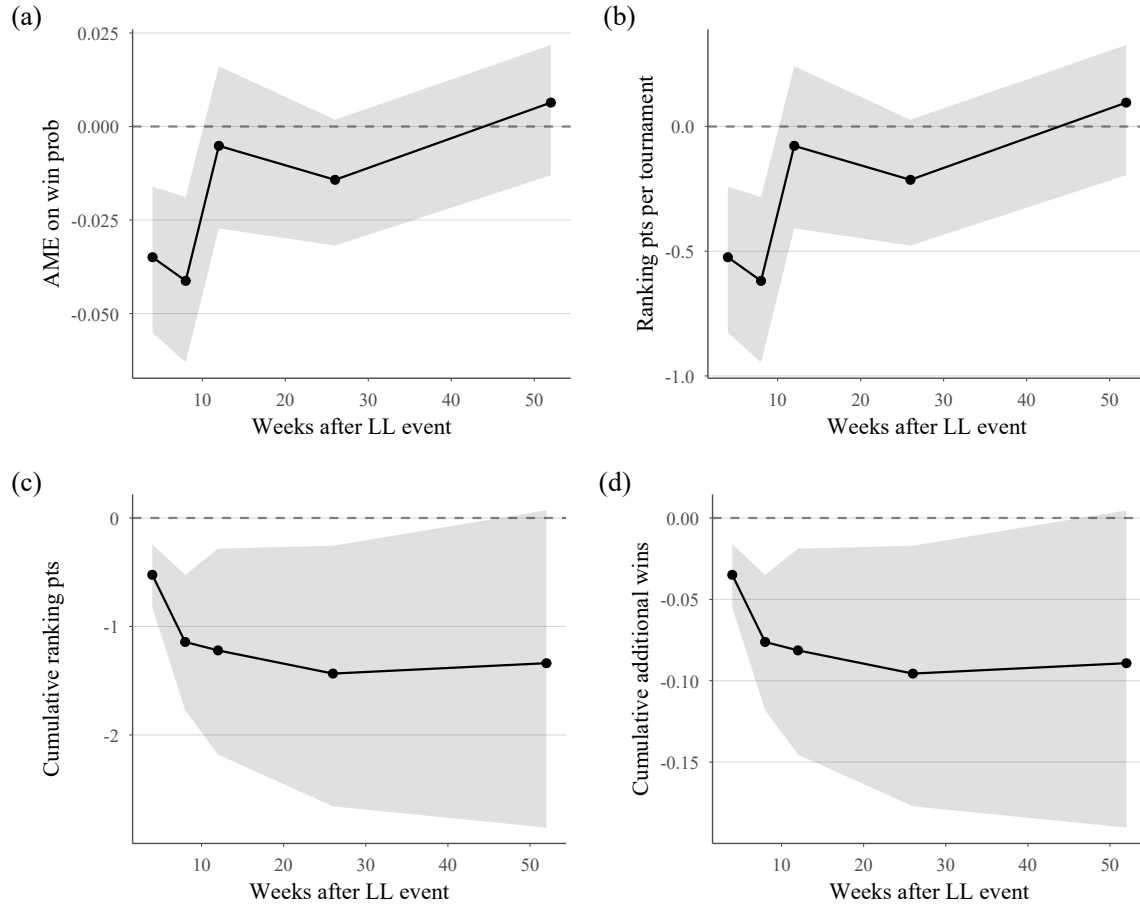


Figure 7: Match-Level Treatment Effects by Calendar Horizon: Non-Grand-Slam WTA

Notes. Same specification as Figure 6. Non-Grand-Slam WTA sample ($N = 80,168$ matches, 747 players). Shaded bands are 95% confidence intervals from player-level block bootstrap (200 replications).

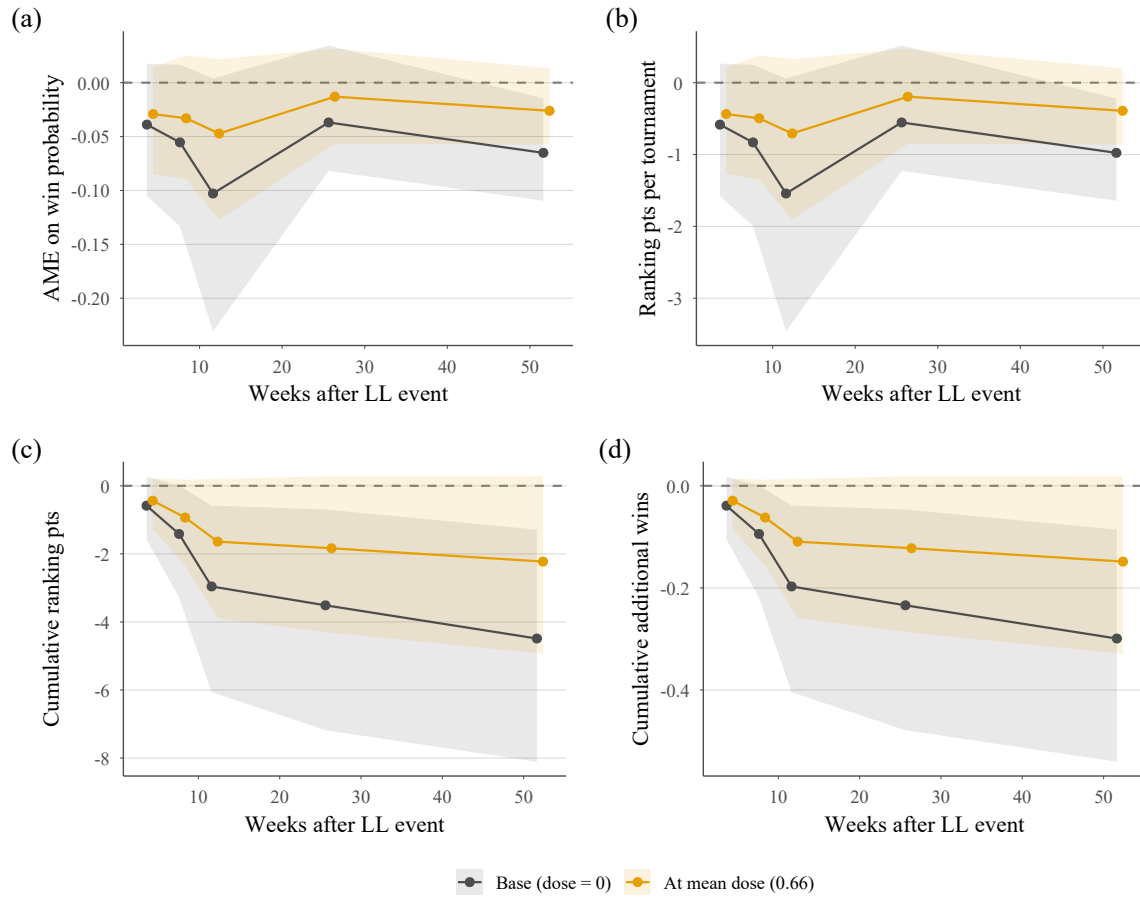


Figure 8: Match-Level Treatment Effects by Horizon and Performance Dose: Grand Slam ATP

Notes. Same four-panel format as Figure 4. Two series: base effect (dose = 0, solid) and effect at mean dose among treated (dashed). Grand Slam ATP sample. Shaded bands are 95% bootstrap CIs.

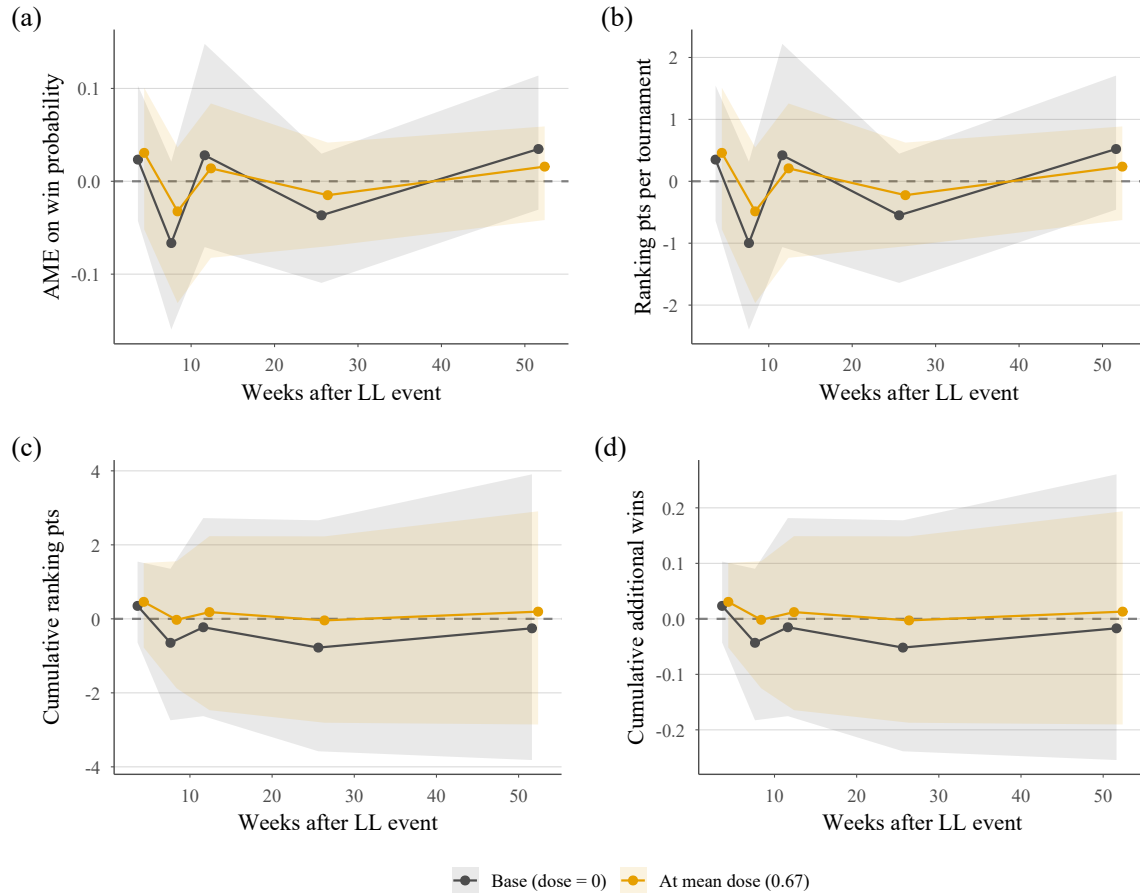


Figure 9: Match-Level Treatment Effects by Horizon and Performance Dose: Grand Slam WTA

Notes. Same specification as Figure 8. Grand Slam WTA sample. Shaded bands are 95% bootstrap CIs.

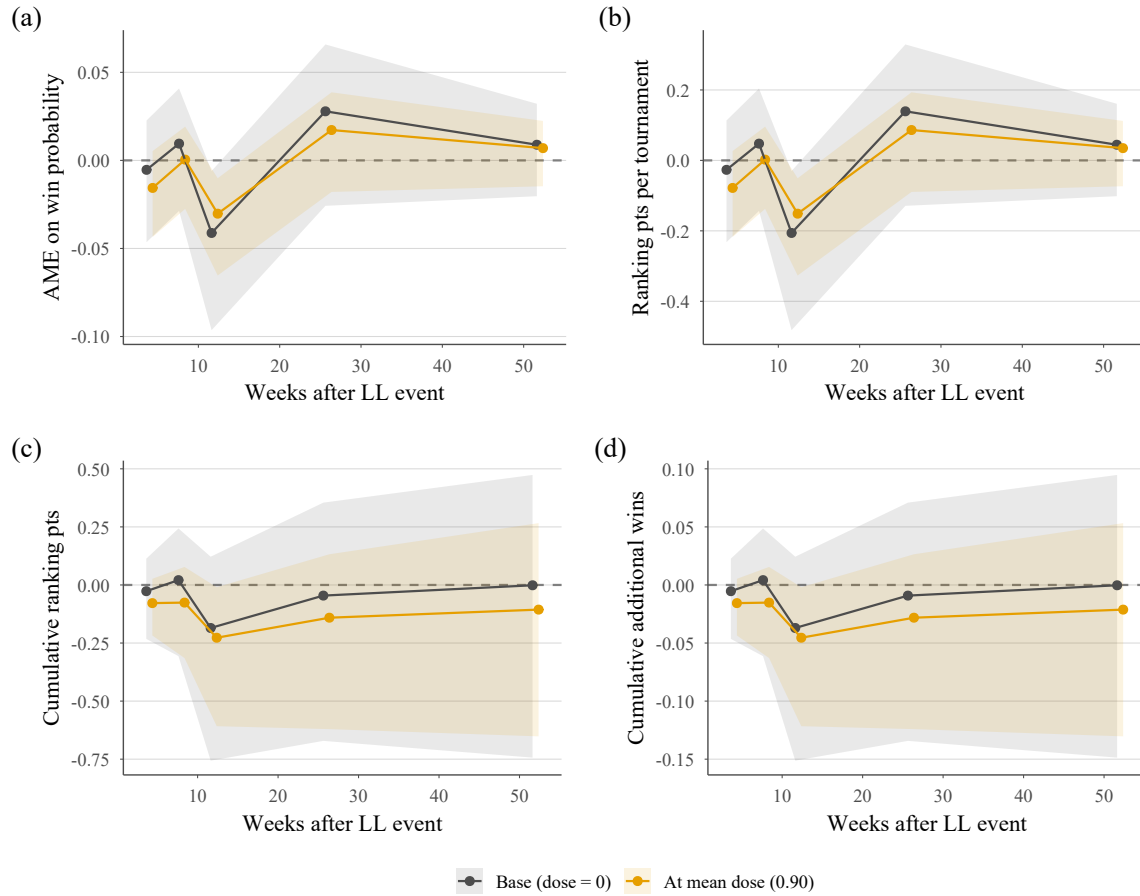


Figure 10: Match-Level Treatment Effects by Horizon and Performance Dose: Non-Grand-Slam ATP

Notes. Same specification as Figure 8, with control function correction for endogenous LL assignment. Non-Grand-Slam ATP sample. Shaded bands are 95% bootstrap CIs.

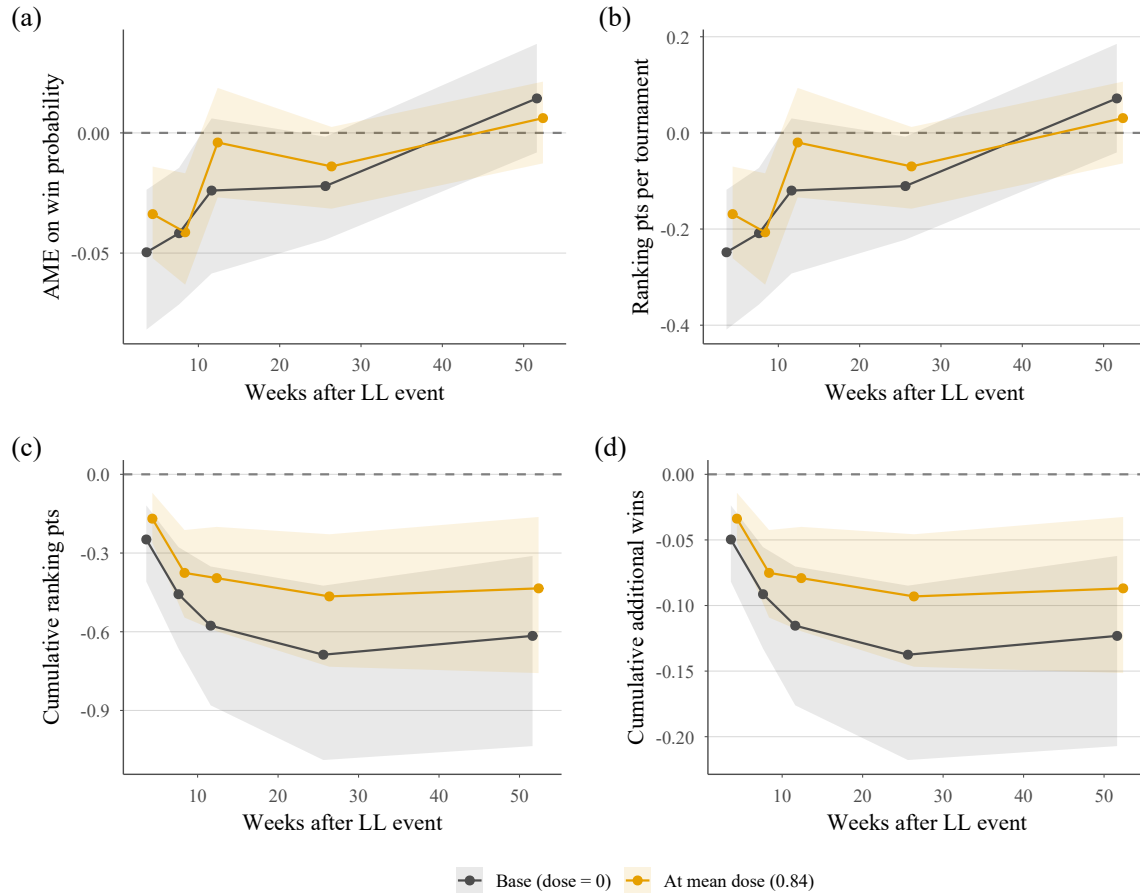


Figure 11: Match-Level Treatment Effects by Horizon and Performance Dose: Non-Grand-Slam WTA

Notes. Same specification as Figure 10. Non-Grand-Slam WTA sample. Shaded bands are 95% bootstrap CIs.

Link to reduced-form results. The GS-ATP estimate ($\hat{\delta} = -0.123$, $p = 0.119$) is consistent with the reduced-form findings in Section 5: positive access effects and null Elo effects. LL entry provides additional tournament appearances through the ranking system, but treated players perform neither better nor worse than controls once opponent strength and context are held constant. The ranking point gains documented in Section 5 come purely from additional tournament participation—more draws entered, more matches played—not from playing better or worse once in the draw. The GS-WTA estimate ($\hat{\delta} = +0.034$, $p = 0.760$) is also a null, reinforcing the access-not-ability interpretation. The WTA dynamic results in Section 5 show marginally significant Elo gains at short horizons; the absence of a corresponding match-level effect suggests that those Elo gains may reflect favorable opponent draws in additional tournaments rather than improved competitive skill per se.

At non-Grand-Slam events, the control-function estimates yield $\hat{\delta} = +0.033$ on the ATP tour (bootstrap SE = 0.050, $p = 0.513$; 113,189 matches from 838 players) and $\hat{\delta} = -0.060$ on the WTA tour (bootstrap SE = 0.032, $p = 0.061$; 80,168 matches from 747 players). The ATP non-GS estimate is small and insignificant, consistent with the Grand Slam null. The WTA non-GS estimate is marginally significant and negative, an unexpected result that may reflect selection: WTA LL recipients at non-GS events who receive their slot by ranking (rather than lottery) may face systematically tougher early-round opponents, producing a mechanical disadvantage in subsequent match outcomes. This estimate should be interpreted with caution and does not alter the overall conclusion that match-level performance effects are generally null across tours and identification strategies. These results are reported in Appendix B.4.

6.4 Reconciling with the Existing Literature

Maity et al. (2025) report positive correlations between lucky loser entry and career outcomes. The Grand Slam tournament performance model, which exploits the lottery for identification, finds no detectable effect of LL entry on match-level win probability on either tour: $\hat{\delta} =$

-0.123 ($p = 0.119$) on the ATP tour and $\hat{\delta} = +0.034$ ($p = 0.760$) on the WTA tour. The null on both tours is consistent with the access-not-ability interpretation: the correlational finding in [Maity et al. \(2025\)](#) likely reflects the additional tournament participation that LL entry generates, not improved competitive performance per se. The causal estimates here establish that LL entry does not detectably change match-level win probability conditional on opponent strength.

The initial conditions literature finds persistent effects of early-career shocks ([Oyer, 2006](#); [Oreopoulos et al., 2012](#)). The present results are consistent with these findings on the access dimension: LL entry causes persistent changes in the number and type of tournaments a player enters. The ability dimension is null on both tours: Elo is flat, and match-level performance conditional on opponent strength is unchanged. This contrast suggests that persistence depends on market transparency: in professional tennis, where performance is measured match by match and rankings update weekly, ability reasserts itself within a year. The access channel alone is sufficient to generate the ranking point gains and tournament entry effects documented in [Section 5](#).

7 Robustness

7.1 Verified Lottery Subsample

The Grand Slam lottery mechanism operates only when withdrawals occur before qualifying completes; post-qualifying withdrawals follow the ranking rule. Withdrawal timing is not recorded in the data, so the full sample may include some ranking-based assignments. To address this contamination concern, I identify events where the selected LL was provably not the highest-ranked candidate, confirming that a lottery determined selection. This yields an ATP verified subsample of $N = 100$ (38 treated, across 34 events) and a WTA verified subsample of $N = 44$ (14 treated).

[Table 20](#) reports results for this verified lottery subsample using the stacked specification.

The ATP verified subsample (100 player-episodes, 38 treated, across 34 confirmed lottery events contributing approximately 500 stacked observations) shows null ranking point effects at all horizons, consistent with limited statistical power in this smaller sample. The ATP verified subsample shows significant *negative* Elo effects at 4 weeks (-17.8 , $p = 0.010$) and 52 weeks (-34.6 , $p = 0.012$). Rather than contradicting the main null Elo finding, this result is consistent with the overseeding mechanism documented in Section 6.3: LL recipients who enter the main draw via confirmed lottery face opponents substantially above their ability level, and their Elo declines as they lose these matches. The main-sample null Elo result averages across both lottery-assigned and potentially ranking-assigned players, where the latter may be better positioned to compete.

The WTA verified subsample (44 player-episodes, 14 treated, contributing approximately 220 stacked observations) shows large positive effects on both ranking points and Elo at all horizons, robust to dropping the three most influential observations. This suggests genuine tour-level heterogeneity: WTA LL recipients may benefit more from main draw experience because the WTA competitive landscape features greater parity and more frequent upsets, creating more room for marginal players to accumulate both points and ability gains.

7.2 First-LL-Only Restriction

The first-LL-only restriction retains only stacked observations from players who have never previously won a Grand Slam LL spot (GS LL won = 0). For the reduced-form dynamic outcomes, the first-LL restriction serves as a robustness check, ensuring that the findings are not driven by players with repeated LL exposure. Most players in the sample experience at most one career-altering opportunity, making the first-LL restriction the closest analog to the labor market “first break” that the initial conditions literature studies.

Tables 21 and 22 report reduced-form dynamic results for this restricted sample. Tournament access effects are qualitatively consistent with the full-sample results.

Table 20: Verified Lottery Subsample: Effect on Career Outcomes

	4w	8w	12w	26w	52w
<i>Panel A: ATP GS</i>					
Ranking points Δ	0.68 (21.81)	-4.56 (21.54)	9.52 (21.48)	43.62 (46.92)	-11.35 (67.57)
Main draws entered	0.16 (0.43)	-0.00 (0.41)	0.23 (0.42)	0.16 (0.59)	0.06 (1.60)
Matches at 250+	-0.11 (0.71)	-0.39 (0.71)	0.04 (0.72)	0.67 (1.15)	0.57 (2.93)
Elo Δ	-18.52** (8.41)	-20.16* (10.30)	-19.68** (9.75)	-6.18 (15.68)	-40.77** (20.29)
<i>N</i>			430		
Player-events			100		
<i>Panel B: WTA GS</i>					
Ranking points Δ	76.57* (42.02)	91.48*** (32.04)	76.71** (37.48)	164.29*** (49.31)	326.77*** (119.83)
Main draws entered	0.48 (0.65)	0.80 (0.70)	1.09 (0.76)	2.16*** (0.80)	5.21** (2.29)
Matches at 250+	0.48 (1.34)	1.34 (1.20)	1.59 (1.26)	4.22*** (1.55)	7.37* (4.47)
Elo Δ	15.34* (7.90)	27.11*** (9.70)	36.25*** (12.90)	43.70*** (14.36)	48.47* (28.38)
<i>N</i>			202		
Player-events			44		
Event FE			Yes		
Horizon FE			Yes		

Notes. Sample restricted to Grand Slam events where the selected LL was not the highest-ranked qualifying loser, confirming lottery-based assignment. ATP: $N = 100$ (38 treated); WTA: $N = 44$ (14 treated). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 21: First-LL-Only ATP Grand Slam: Dynamic Effects

	4w	8w	12w	26w	52w
Ranking Points Δ	8.22 (37.73)	2.68 (38.24)	15.77 (38.94)	43.73 (47.90)	-59.71 (84.22)
Elo Δ	-10.10 (24.14)	-11.27 (26.59)	-23.59 (25.82)	-28.21 (27.13)	-65.13 (44.29)
Main Draws	-3.09** (1.22)	-2.86** (1.21)	-2.59** (1.19)	-1.97 (1.35)	-1.33 (2.10)
Matches (250+)	-4.54** (2.00)	-4.03** (1.93)	-3.67* (2.02)	-2.60 (2.18)	-2.84 (4.18)
N	206				
Z^{pre} controls	Yes				

Notes. Restricted to stacked observations from ATP players with zero prior Grand Slam LL wins, 2006–2024. Event FE specification with pre-treatment controls. Standard errors clustered at the player level.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 22: First-LL-Only WTA Grand Slam: Dynamic Effects

	4w	8w	12w	26w	52w
Ranking Points Δ	2.71 (35.82)	4.48 (30.21)	-38.34 (25.96)	1.81 (32.43)	137.29 (87.05)
Elo Δ	7.89 (15.05)	12.05 (15.98)	18.44 (16.24)	33.13* (19.17)	41.52 (32.52)
Main Draws	-0.15 (0.88)	-0.20 (0.92)	-0.38 (0.91)	-0.60 (1.08)	0.39 (2.23)
Matches (250+)	0.60 (0.89)	0.84 (0.78)	0.34 (0.75)	0.42 (0.70)	1.15 (2.62)
N	202				
Z^{pre} controls	Yes				

Notes. Restricted to stacked observations from WTA players with zero prior Grand Slam LL wins, 2006–2024. Event FE specification with pre-treatment controls. Standard errors clustered at the player level.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7.3 Full Estimation Sample: Tournament Performance

The match-level tournament performance results in Section 6.3 use all LL candidates as the primary specification. The estimates—GS-ATP $\hat{\delta} = -0.123$ (bootstrap SE = 0.079, $p = 0.119$; 8,543 matches from 130 players) and GS-WTA $\hat{\delta} = +0.034$ (bootstrap SE = 0.112, $p = 0.760$; 5,391 matches from 98 players)—are null on both tours. Four-specification robustness results are reported in Appendix Tables 37 and 38.

7.4 Pre-Treatment Balance

Table 23 reports formal balance tests on the full set of pre-treatment covariates Z_{ie}^{pre} : ranking points / 1000 (linear and quadratic), Elo / 100 (linear and quadratic), surface Elo / 100 (linear and quadratic), age, prior LL history (GS won, GS not won, non-GS won, non-GS not won), and a had-prior-LL indicator. In the ATP Grand Slam sample ($N = 248$), the joint F -test yields $p = 0.230$, consistent with randomization. In the WTA sample ($N = 132$), the joint F -test yields $p = 0.333$. The balance tests are conducted at the player-episode level (one observation per player per Grand Slam appearance). Stacking across horizons mechanically multiplies each observation five times, inflating the F -statistic without adding independent variation; the appropriate balance test is therefore at the player-episode level. The inclusion of Z_{ie}^{pre} in all dynamic specifications absorbs any residual finite-sample imbalance. Event study plots (Figures 1 and 2) show parallel trajectories in the weeks preceding each Grand Slam, confirming the absence of pre-treatment divergence.

Table 47 reports the corresponding balance tests for the non-Grand-Slam samples. As expected for the non-randomized non-GS setting, treated players have significantly higher ranking points, Elo, and more prior LL experience (joint F -test: $p < 0.001$ for both tours). This imbalance motivates the control function approach described in Section 4.5.

Table 23: Pre-Treatment Balance: Grand Slam Samples

Variable	ATP			WTA		
	LL	Control	p	LL	Control	p
Ranking points / 1000	0.49	0.47	0.087	0.56	0.48	0.003
(Ranking points / 1000) ²	0.25	0.23	0.106	0.34	0.25	0.003
Elo / 100	0.18	0.17	0.532	0.19	0.19	0.030
(Elo / 100) ²	0.03	0.03	0.524	0.04	0.04	0.031
Surface Elo / 100	16.53	16.59	0.685	17.44	17.44	0.993
(Surface Elo / 100) ²	274.60	276.78	0.678	307.60	306.83	0.945
Age	26.89	27.08	0.690	24.53	24.16	0.593
Had prior LL	0.54	0.49	0.467	0.39	0.37	0.844
Prior GS LL won	0.30	0.25	0.477	0.15	0.09	0.344
Prior GS LL not won	1.11	0.88	0.143	0.39	0.37	0.885
Prior non-GS LL won	1.15	0.74	0.023	0.65	0.53	0.529
Prior non-GS LL not won	1.97	1.69	0.302	1.81	1.27	0.145
Joint F -test p -value	0.230			0.333		
N	248			132		

Notes. Columns compare means for treated (LL selected) and control (unselected) players at the player-episode level. Covariates include the full Z_{ie}^{pre} specification: ranking points / 1000 (linear and quadratic), Elo / 100 (linear and quadratic), surface Elo / 100 (linear and quadratic), age, and prior LL history indicators. Joint F -statistic tests the null that all covariates are jointly balanced. ATP: $N = 248$ player-episodes (132 treated, 116 control); WTA: $N = 132$ (54 treated, 78 control). Grand Slam sample, 2006–2024.

8 Conclusion

A single randomly assigned tournament opportunity in professional tennis causes gains in tournament access but does not uniformly improve playing ability. In the full ATP Grand Slam sample ($N = 248$), a stacked specification shows that lottery selection increases ranking points by 31.9 at 4 weeks ($p = 0.028$), 38.5 at 12 weeks ($p = 0.007$), and 49.2 at 26 weeks ($p = 0.054$), with main draw entries increasing by 1.1 over 26 weeks ($p = 0.006$). On the WTA tour ($N = 132$), effects are larger at short horizons—64.4 points at 4 weeks ($p = 0.003$) and 78.7 at 8 weeks ($p < 0.001$)—but fade by 26 weeks. At the immediate event, treated players are 32 percentage points more likely to win a main draw match, earning an average of 29.66 ranking points. Elo ratings—which adjust for opponent quality—are unchanged at every horizon on both tours. A match-level tournament performance model finds no significant effect on match win probability on either tour: $\hat{\delta} = -0.123$ ($p = 0.119$; 8,543 matches from 130 players) on the ATP tour and $\hat{\delta} = +0.034$ ($p = 0.760$; 5,391 matches from 98 players) on the WTA tour. Ranking point gains come from additional participation rather than improved match performance. The full estimation sample (including repeat LL candidates) produces qualitatively similar results, with expected attenuation. The non-Grand-Slam control function estimates corroborate these findings: ATP ranking point gains of 12.2 at 4 weeks ($p = 0.065$) and 28.1 at 26 weeks ($p = 0.009$), with significant tournament access gains at longer horizons.

The dominant mechanism is access accumulation. The ranking system does not adjust for opponent quality: a first-round win at a Grand Slam earns the same points regardless of the opponent’s strength. When a lucky loser earns points at the focal event, their improved ranking grants entry to subsequent tournaments, generating further opportunities to accumulate points. Match-level performance conditional on opponent strength is unchanged on both tours: treated players perform neither better nor worse than controls once in the draw. Ranking point gains on both tours come from more draws entered and more matches played, not from improved competitive ability.

These findings have implications for any labor market where threshold-based access generates cumulative advantage. Second-chance mechanisms—wildcard entries in sports, waitlist admissions in universities, trial hiring periods in firms—can provide real access benefits to marginal candidates. But when performance is continuously measured and positions are regularly updated, the signal from sustained ability dominates the noise from a single favorable shock. The contrast with [Oyer \(2006\)](#) and [Oreopoulos et al. \(2012\)](#), where initial conditions cause persistent career effects in less transparent markets, suggests that persistence depends on how transparently performance is measured and aggregated.

Several limitations apply. First, the lottery contamination caveat—that an unknown fraction of observations were assigned by ranking rather than by draw—biases toward attenuation but cannot be fully resolved without withdrawal timing data; the verified lottery subsample and first-LL-only restriction mitigate this concern. Second, the WTA Grand Slam sample ($N = 132$) has limited power, and WTA dynamic effects should be interpreted with caution. Third, the non-Grand-Slam control function estimates rely on the exact enumeration selection probability P_i^{LL} for identification and may not generalize to all qualifying losers. Fourth, I measure tournament access and match performance, which are imperfect proxies for the career outcomes that matter most to players: earnings, career length, and sponsorship. Future work could examine longer-run outcomes—career length, total earnings, peak ranking achieved—to determine whether the temporary access boost has lasting consequences through sponsorship or confidence channels. The finding that access shocks generate ranking gains without affecting match-level ability deserves study in other settings with lottery-based or threshold-based access to test the generality of the result.

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A Additional Theoretical and Econometric Details

This appendix provides details on the tournament performance model (Section 6.3), including the control-function approach used for non-Grand-Slam events, and additional tables for the non-Grand-Slam control function analysis.

A.1 Tournament Trajectory Likelihood

Let player i enter tournament t and face a sequence of opponents indexed by match $m = 1, 2, \dots$. Define the binary outcome:

$$w_{itm} = \mathbf{1}[\text{player } i \text{ wins match } m \text{ at tournament } t] \quad (8)$$

Define the stopping time:

$$T_{it} = \min\{m : w_{itm} = 0\} \quad (9)$$

with the convention that if the player wins the tournament, $T_{it} = R_t + 1$, where R_t is the total number of rounds.

The tournament trajectory is:

$$F_{it} = (w_{it1}, w_{it2}, \dots, w_{it,T_{it}}) \quad (10)$$

Let the match-level win probability be:

$$p_{itm} = P(w_{itm} = 1 \mid X_{itm}, D_{ie}, v_{ie}) \quad (11)$$

Under conditional independence across matches given covariates:

$$P(F_{it}) = \left(\prod_{m=1}^{T_{it}-1} p_{itm} \right) \times (1 - p_{it,T_{it}}) \quad (12)$$

If the player wins all rounds:

$$P(F_{it}) = \prod_{m=1}^{R_t} p_{itm} \quad (13)$$

Taking logs:

$$\log P(F_{it}) = \sum_{m=1}^{T_{it}} [w_{itm} \log p_{itm} + (1 - w_{itm}) \log(1 - p_{itm})] \quad (14)$$

This establishes that the likelihood factorizes exactly into match-level Bernoulli contributions.

A.2 Connection to an Inhomogeneous Geometric Distribution

The stopping time T_{it} follows an inhomogeneous geometric distribution:

$$P(T_{it} = m) = \left(\prod_{s=1}^{m-1} p_{its} \right) (1 - p_{itm}) \quad (15)$$

with success probabilities varying across matches. This differs from the standard geometric distribution due to heterogeneity in opponent strength and match context.

The expected number of wins is:

$$E[W_{it}] = \sum_{r=1}^{R_t} P(T_{it} > r) = \sum_{r=1}^{R_t} \prod_{s=1}^r p_{its} \quad (16)$$

This identity underlies the simulation-based estimators used in the main text.

A.3 Match-Level Model and Likelihood Reduction

The match-level probability is specified as:

$$p_{itm} = \Lambda(\beta' X_{ijm} + \gamma' Z_{ie}^{pre} + \delta D_{ie} + \rho v_{ie}) \quad (17)$$

where X_{ijm} contains the match-level pairwise covariates (ranking points difference / 1000 linear and quadratic, Elo difference / 100 linear and quadratic, surface Elo difference / 100 linear and quadratic, Laplace-smoothed H2H win proportion, H2H encounter count, age difference, and clay and grass surface indicators) and Z_{ie}^{pre} contains the pre-treatment focal player characteristics (ranking points / 1000 linear and quadratic, Elo / 100 linear and quadratic, surface Elo / 100 linear and quadratic, and player age).

Substituting into the trajectory likelihood and using the log decomposition yields:

$$\mathcal{L} = \sum_{i,t,m} [w_{itm} \log \Lambda(z_{itm}) + (1 - w_{itm}) \log(1 - \Lambda(z_{itm}))] \quad (18)$$

where $z_{itm} = \beta' X_{ijm} + \gamma' Z_{ie}^{pre} + \delta D_{ie} + \rho v_{ie}$.

Thus, estimation reduces to a standard logit on the stacked match-level data.

A.4 Control Function Approach

The control function corrects for endogenous LL assignment at non-Grand-Slam events. The selection probability P_i^{LL} is constructed from predicted match-winning probabilities obtained from a weighted logit model estimated on all qualifying round matches (Section A.5), combined via exact enumeration of all final-round outcome combinations. Under the maintained assumption that the latent selection error and the outcome error are jointly normal:

$$(u_{ie}, \varepsilon_{itm}) \sim \mathcal{N}(0, \Sigma) \quad (19)$$

the generalized residual is computed analytically from P_i^{LL} :

$$\hat{v}_{ie} = D_{ie} \cdot \frac{\phi(\Phi^{-1}(P_i^{LL}))}{P_i^{LL}} - (1 - D_{ie}) \cdot \frac{\phi(\Phi^{-1}(P_i^{LL}))}{1 - P_i^{LL}} \quad (20)$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are the standard normal PDF and CDF. Including $\hat{v}_{ie}$ in the outcome equation corrects for selection on unobservables. The coefficient ρ on $\hat{v}_{ie}$ tests for endogeneity:

if $\rho \neq 0$, LL assignment is correlated with unobserved determinants of the outcome.

A.5 Selection Probability Construction

The selection probability underlying the non-Grand-Slam control function and the tournament performance model is computed from the full combinatorial structure of qualifying match pairings. For each player i at event e with N_e players in the final round of qualifying and c_e LL slots available, the selection probability is:

$$P_i^{LL} = \Pr(\text{rank among losers} \leq c_e \mid i \text{ loses})$$

Construction proceeds in two steps. First, a weighted logit model of match outcomes is estimated on qualifying round matches at non-Grand-Slam LL-granting events, separately for each tour (ATP: $N = 9,775$; WTA: $N = 10,199$ qualifying matches). The match-level covariates X_{ijm} include ranking points difference / 1000 (linear and quadratic), Elo difference / 100 (linear and quadratic), surface Elo difference / 100 (linear and quadratic), Laplace-smoothed head-to-head win proportion, head-to-head encounter count, and age difference, along with event-level surface indicators (clay and grass). The logit is estimated by weighted MLE, with weights equal to the total points in play (sum of service points) in each match; qualifying round number serves as a fallback weight when service point data are unavailable. Estimating on all qualifying rounds—not just the final round—improves parameter precision by exploiting the full set of qualifying matches. The same logit specification is used for the selection probability, the tournament performance model, and the treatment dose analysis. This yields predicted match-winning probabilities $\hat{P}(a_m \rightarrow b_m \mid X, Z)$ for each final-round qualifying match m between players a_m and b_m ; players with missing pre-match rankings are assigned a default predicted win probability of 0.5.

Second, given these predicted probabilities and the $N_e - 1$ other final-round qualifying matches (excluding player i 's match, since i is conditioned on having lost), the probability

is computed by exact enumeration over all 2^{N_e-1} outcome combinations. Each combination has probability equal to the product of the individual match outcome probabilities from the logit (invoking conditional independence of match outcomes given observables). For each combination, the set of losers is determined and their rankings compared; P_i^{LL} is the sum of probabilities of all combinations in which i 's rank falls within the top c_e among losers. When $N_e - 1$ exceeds 20 (making exact enumeration computationally prohibitive), the probability is approximated by Monte Carlo simulation with 10,000 draws; this applies to 115 of 5,208 observations. The resulting P_i^{LL} is systematically lower than a naive Bernoulli convolution that treats qualifying outcomes as independent coin flips without accounting for the match pairing structure (mean 0.279 vs. 0.506 across the non-GS sample). The difference arises because exact enumeration correctly conditions on the specific draw structure—which players are paired against each other—rather than treating each player's loss probability as independent of the draw.

A.6 Simulation-Based Counterfactuals

Define counterfactual probabilities:

$$p_{itm}(d) = \Lambda(X'_{itm}\hat{\beta} + \hat{\delta} \cdot d) \quad (21)$$

Expected wins:

$$E[W_{it}(d)] = \sum_{r=1}^{R_t} \prod_{s=1}^r p_{its}(d) \quad (22)$$

Tournament win probability:

$$\pi_{it}(d) = \prod_{r=1}^{R_t} p_{itr}(d) \quad (23)$$

Average marginal effect:

$$AME = \frac{1}{N} \sum_{i,t,m} [p_{itm}(1) - p_{itm}(0)] \quad (24)$$

All quantities are computed for every player-tournament observation in the data and then averaged, integrating over actual draw compositions and opponent strengths.

A.7 Bootstrap Inference

Inference for the tournament performance model is conducted via player-level block bootstrap:

1. Resample players i with replacement.
2. Retain full histories across all events and matches for each sampled player.
3. Re-estimate the outcome equation with a fresh draw of player clusters.
4. Recompute all counterfactual quantities (AME, expected wins, tournament win probability).

This preserves within-player dependence across tournaments and matches. Standard errors and confidence intervals are computed from the bootstrap distribution using 200 replications.

A.8 Additional Identification Discussion

Identification of the tournament performance model relies on three conditions:

- *Exogeneity of the withdrawal process conditional on rank ordering.* Main draw withdrawals must be independent of the specific identity of qualifying losers. This is plausible because withdrawals are driven by injury or scheduling among main draw players, who are ranked well above the qualifying losers.

- *Independence of P_i^{LL} from match-level shocks ε_{itm} .* The selection probability depends on the qualifying match outcomes of other players, which are independent of the focal player's future match-level shocks.
- *Correct specification of the selection probability.* The exact enumeration correctly models the probability that player i 's rank falls within the top c among losers given the draw structure. The joint normality of $(u_{ie}, \varepsilon_{itm})$ is a maintained assumption. Misspecification of either could affect the control function correction and bias $\hat{\delta}$.

Potential violations arise if withdrawals correlate with unobserved player-specific shocks (e.g., if a main draw player withdraws because they know they will face a particular qualifying loser). The institutional structure makes this unlikely: main draw withdrawals are typically announced before the draw is made, and the identity of potential replacements is not known to withdrawing players.

B Non-Grand-Slam Tables

Table 24 reports the distribution of LL candidates across tournament tiers, with ATP events disaggregated into Masters 1000 (862 candidates), ATP 500 (594), and ATP 250 (1,216).

Tournament Level	ATP			WTA		
	LL	Control	Total	LL	Control	Total
Masters 1000 / WTA 1000	188	674	862	58	237	295
ATP 500 / WTA 500	203	391	594	332	808	1140
ATP 250 / WTA 250	471	745	1216	365	745	1110
Total	862	1810	2672	755	1790	2545

Notes. Sample period: 2007–2024 for ATP non-Grand-Slam events; 2009–2024 for WTA non-Grand-Slam events. Candidates are final-round qualifying losers at events where at least one LL slot was awarded. ATP tiers: Masters 1000, ATP 500, ATP 250. WTA tiers: WTA 1000, WTA 500, WTA 250.

Table 25 cross-tabulates LL candidacies and wins for the non-Grand-Slam sample.

Opportunities	ATP				WTA			
	Won=0	Won=1	Won=2	Won=3+	Won=0	Won=1	Won=2	Won=3+
1	266	74	0	0	216	52	0	0
2	81	65	13	0	62	54	10	0
3	46	34	21	2	35	49	9	0
4	13	25	17	3	24	22	17	11
5+	14	32	53	103	18	38	62	73
Total	420	230	104	108	355	215	98	84

Notes. Same structure as Table 2 for non-Grand-Slam events. Sample period: ATP 2007–2024, WTA 2009–2024.

Tables 26 and 27 report summary statistics for the non-Grand-Slam samples on the ATP and WTA tours, respectively.

B.1 ATP Non-Grand-Slam Results

Table 28 reports the non-Grand-Slam control function immediate effects. LL recipients at non-GS events earn additional ranking points at the focal event on both tours.

Table 26: Summary Statistics: ATP Non-Grand-Slam Sample

Variable	LL Selected			Not Selected		
	Mean	SD	N	Mean	SD	N
<i>Panel A: Pre-Treatment</i>						
Age	27.06	4.07	874	26.46	4.38	1834
Ranking points	511.36	237.56	874	356.29	223.72	1834
Elo rating	1749.04	104.87	874	1685.80	115.43	1834
Surface Elo	1701.40	124.45	874	1666.89	120.71	1834
Prior GS LL won	0.17	0.44	874	0.13	0.43	1834
Prior GS LL not won	0.74	1.08	874	0.70	1.11	1834
Prior non-GS LL won	0.96	1.40	874	0.79	1.29	1834
Prior non-GS LL not won	1.66	2.00	874	1.37	1.76	1834
Had prior LL	0.54	0.50	874	0.43	0.49	1834
P_i^{LL}	0.72	0.35	874	0.13	0.26	1834
<i>Panel B: Outcomes</i>						
Ranking points Δ (4w)	14.77	74.29	784	1.09	45.97	1737
Ranking points Δ (26w)	28.63	231.62	552	8.86	158.15	1347
Ranking points Δ (52w)	16.90	339.05	403	13.26	247.52	1043
Elo Δ (4w)	0.82	26.99	741	2.53	26.55	1571
Elo Δ (26w)	7.43	60.87	474	8.64	60.87	1096
Elo Δ (52w)	4.20	77.74	343	12.62	77.81	840
Main draws entered (26w)	5.32	4.97	594	2.76	3.78	1416
Matches at 250+ events (26w)	8.87	9.38	594	4.29	6.86	1416

Notes. ATP non-Grand-Slam sample at events with at least one LL slot awarded, 2007–2024. $N = 2,708$ (874 treated). Pre-treatment controls include ranking points (linear and quadratic), Elo rating (linear and quadratic), prior GS LL spots won, prior GS LL candidacies not won, prior non-GS LL spots won, prior non-GS LL candidacies not won, and player age.

Table 27: Summary Statistics: WTA Non-Grand-Slam Sample

Variable	LL Selected			Not Selected		
	Mean	SD	N	Mean	SD	N
<i>Panel A: Pre-Treatment</i>						
Age	25.35	4.12	755	24.91	4.39	1788
Ranking points	636.98	327.34	755	454.88	272.46	1790
Elo rating	1937.83	114.22	755	1883.85	127.32	1790
Surface Elo	1845.69	142.65	755	1818.84	139.30	1790
Prior GS LL won	0.10	0.35	755	0.08	0.31	1790
Prior GS LL not won	0.41	0.66	755	0.36	0.67	1790
Prior non-GS LL won	0.96	1.43	755	0.70	1.18	1790
Prior non-GS LL not won	1.78	1.95	755	1.65	2.02	1790
Had prior LL	0.50	0.50	755	0.42	0.49	1790
P_i^{LL}	0.71	0.35	755	0.12	0.25	1790
<i>Panel B: Outcomes</i>						
Ranking points Δ (4w)	15.50	98.25	699	6.45	65.96	1690
Ranking points Δ (26w)	4.35	290.11	482	26.39	203.38	1230
Ranking points Δ (52w)	8.05	488.98	345	27.74	322.45	898
Elo Δ (4w)	1.79	25.82	675	1.17	26.19	1607
Elo Δ (26w)	5.73	59.42	417	10.00	61.08	1013
Elo Δ (52w)	12.66	77.39	298	14.18	78.44	721
Main draws entered (26w)	5.08	4.37	532	3.21	3.71	1320
Matches at 250+ events (26w)	8.30	8.44	532	5.14	6.85	1320

Notes. WTA non-Grand-Slam sample at events with at least one LL slot awarded, 2009–2024. $N = 2,545$ (755 treated). Pre-treatment controls include ranking points (linear and quadratic), Elo rating (linear and quadratic), prior GS LL spots won, prior GS LL candidacies not won, prior non-GS LL spots won, prior non-GS LL candidacies not won, and player age.

Table 28: ATP Non-Grand-Slam: Immediate Event-Level Effects (Control Function)

Outcome	<i>t</i> -test	LOO-IV	
	LL Mean	$\hat{\beta}_{IV}$	(SE)
<i>ATP</i>			
Main-draw match win	0.35***	0.31***	(0.02)
Main-draw matches played	1.53***	1.46***	(0.04)
Ranking points at event	151.27***	137.64***	(8.96)
<i>N</i>	874 / 1834	2708	
<i>WTA</i>			
Main-draw match win	0.37***	0.33***	(0.02)
Main-draw matches played	1.50***	1.45***	(0.03)
Ranking points at event	135.16***	131.39***	(6.60)
<i>N</i>	755 / 1790	2545	
<i>Pooled</i>			
Main-draw match win	0.36***	0.32***	(0.02)
Main-draw matches played	1.52***	1.45***	(0.03)
Ranking points at event	143.80***	134.21***	(5.71)
<i>N</i>	1629 / 3624	5253	

Notes. Control function estimates using the generalized residual from the selection probability P_i^{LL} (exact enumeration). ATP non-Grand-Slam sample ($N = 2,708$), 2007–2024. All specifications control for pre-treatment ranking points (linear and quadratic), Elo / 100 (linear and quadratic), prior GS LL spots won, prior GS LL candidacies not won, prior non-GS LL spots won, prior non-GS LL candidacies not won, age, and event fixed effects. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 29 reports the ATP non-Grand-Slam CF dynamic results.

Table 29: ATP Non-Grand-Slam: Dynamic Effects (Stacked, Control Function)

	4w	8w	12w	26w	52w
Ranking Pts Δ	12.2* (6.6)	11.0 (7.2)	6.0 (8.0)	28.1*** (10.7)	6.5 (22.8)
Elo Δ	1.1 (3.7)	3.3 (3.9)	2.6 (4.2)	8.1 (7.9)	9.0 (13.3)
Main Draws	-1.5*** (0.2)	-1.0*** (0.1)	-0.4*** (0.1)	1.7*** (0.2)	5.0*** (0.6)
Matches 250+	-2.8*** (0.3)	-1.9*** (0.3)	-0.9*** (0.2)	2.8*** (0.4)	8.6*** (1.0)
Win %	-0.05* (0.02)	-0.03*** (0.01)	-0.01 (0.02)	-0.01 (0.02)	-0.01 (0.02)

Notes. Control function estimates using the generalized residual from P_i^{LL} . ATP non-Grand-Slam sample ($N = 2,708$), 2007–2024. All specifications control for pre-treatment ranking points (linear and quadratic), Elo / 100 (linear and quadratic), prior GS LL spots won, prior GS LL candidacies not won, prior non-GS LL spots won, prior non-GS LL candidacies not won, age, and event fixed effects. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 30 reports ATP non-Grand-Slam heterogeneity results.

Tables 31 and 34 restrict the non-GS CF sample to players with no prior LL wins. The ATP first-LL subsample (1,424 player-episodes, 397 treated) shows persistent ranking point effects across all horizons and significant access gains at 26 and 52 weeks. The WTA first-LL subsample (1,423 player-episodes, 378 treated) shows weaker ranking point effects but significant access gains emerging by 52 weeks.

B.2 WTA Non-Grand-Slam Results

Tables 32–34 report the WTA non-Grand-Slam control function results, which mirror the ATP non-Grand-Slam tables above.

Table 30: ATP Non-Grand-Slam: Heterogeneous Effects (Control Function)

	4w		8w		12w		26w		
	β_h	$\beta_h \times X_k$	β_h	$\beta_h \times X_k$	β_h	$\beta_h \times X_k$	β_h	$\beta_h \times X_k$	β_h
<i>Interaction: Above-Median Rank</i>									
Ranking Points Δ	9.14 (7.36)	4.87 (9.94)	8.88 (7.79)	4.40 (10.35)	-0.79 (8.25)	10.10 (11.03)	17.40 (12.77)	10.30 (17.36)	12.4 (18.5)
Elo Δ	0.25 (3.15)	-0.63 (3.69)	0.02 (3.54)	-0.12 (4.24)	-2.03 (4.06)	0.91 (4.69)	6.43 (5.82)	-8.67 (6.57)	4.8 (8.4)
Main Draws	0.67*** (0.13)	-2.73*** (0.18)	0.46*** (0.13)	-1.76*** (0.18)	0.19 (0.13)	-0.60*** (0.18)	-0.66*** (0.19)	3.21*** (0.28)	-1.18 (0.4)
Matches (250+)	1.10*** (0.23)	-4.93*** (0.34)	0.72*** (0.23)	-3.19*** (0.34)	0.17 (0.24)	-1.11*** (0.34)	-1.25*** (0.34)	5.63*** (0.53)	-2.49 (0.7)
Win %	0.03 (0.02)	-0.06** (0.02)	0.01 (0.02)	-0.04* (0.02)	0.00 (0.02)	-0.03 (0.02)	0.02 (0.01)	-0.04*** (0.02)	0.0 (0.0)
<i>Interaction: Above-Median Age</i>									
Ranking Points Δ	-0.69 (7.97)	25.29*** (7.84)	5.86 (8.89)	12.19 (9.27)	5.15 (10.01)	2.41 (10.80)	43.99*** (14.27)	-37.34** (17.94)	65.73 (25.6)
Elo Δ	-5.09* (2.85)	10.31*** (3.17)	-1.40 (3.33)	3.70 (3.75)	-2.18 (3.67)	2.79 (4.04)	8.12* (4.93)	-14.83** (5.79)	15.64 (6.8)
Main Draws	-1.44*** (0.18)	0.10 (0.20)	-0.96*** (0.16)	0.12 (0.17)	-0.39** (0.15)	0.14 (0.16)	1.58*** (0.26)	-0.05 (0.31)	5.02* (0.6)
Matches (250+)	-2.84*** (0.33)	0.40 (0.39)	-1.89*** (0.30)	0.29 (0.34)	-0.89*** (0.30)	0.22 (0.32)	2.85*** (0.52)	-0.40 (0.60)	9.10* (1.2)
Win %	-0.01 (0.02)	0.00 (0.02)	-0.02 (0.01)	0.00 (0.02)	-0.02* (0.01)	0.01 (0.02)	-0.01 (0.01)	0.00 (0.01)	-0.0 (0.0)
<i>Interaction: Had Prior LL</i>									
Ranking Points Δ	2.81 (7.63)	17.80** (7.71)	7.31 (8.65)	8.57 (9.53)	5.36 (9.93)	1.87 (11.48)	36.88** (14.74)	-24.11 (18.82)	19.6 (24.5)
Elo Δ	-0.82 (2.86)	1.32 (3.18)	1.72 (3.26)	-3.00 (3.63)	1.24 (3.68)	-4.48 (3.96)	9.77** (4.93)	-18.25*** (5.74)	5.8 (7.0)
Main Draws	-0.96*** (0.16)	-0.79*** (0.19)	-0.61*** (0.15)	-0.52*** (0.17)	-0.22 (0.14)	-0.18 (0.16)	1.20*** (0.24)	0.65** (0.30)	3.85* (0.5)
Matches (250+)	-1.93*** (0.31)	-1.27*** (0.36)	-1.27*** (0.28)	-0.85** (0.33)	-0.53* (0.28)	-0.44 (0.31)	2.23*** (0.47)	0.76 (0.60)	6.81* (1.1)
Win %	0.00 (0.02)	-0.02 (0.02)	0.01 (0.01)	-0.04** (0.02)	-0.00 (0.01)	-0.03* (0.02)	0.01 (0.01)	-0.04*** (0.01)	0.0 (0.0)
Z^{pre} controls					Yes (main effects)				

Notes. Control function estimates by subgroup. ATP non-Grand-Slam sample, 2007–2024. Each row reports a set of estimates. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 31: ATP Non-Grand-Slam: First LL Episode Only (Control Function)

	4w	8w	12w	26w	52w
Ranking Points Δ	21.84 (13.77)	22.64* (13.31)	16.16 (13.32)	44.62** (18.74)	14.82 (31.40)
Elo Δ	16.19** (6.93)	17.03** (7.09)	15.17** (7.06)	22.23*** (8.41)	15.14 (11.24)
Main Draws	-1.13*** (0.29)	-0.69** (0.27)	-0.13 (0.26)	1.36*** (0.36)	3.91*** (0.83)
Matches (250+)	-2.01*** (0.58)	-1.30** (0.51)	-0.20 (0.46)	2.49*** (0.63)	6.99*** (1.60)
N	2,751				
Z^{pre} controls	Yes				

Notes. Control function estimates restricted to each player's first LL candidacy episode. ATP non-Grand-Slam sample ($N = 1,424$, 397 treated), 2007–2024. Stacked specification with horizon-specific treatment effects. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 32: WTA Non-Grand-Slam: Dynamic Effects (Stacked, Control Function)

	4w	8w	12w	26w	52w
Ranking Pts Δ	27.4*** (9.4)	22.2** (10.1)	23.0** (11.3)	3.1 (15.6)	4.7 (31.0)
Elo Δ	6.6*** (2.5)	5.5** (2.8)	8.9*** (2.9)	3.1 (4.4)	2.0 (5.9)
Main Draws	-0.9*** (0.1)	-0.5*** (0.1)	-0.1 (0.1)	1.2*** (0.2)	3.2*** (0.4)
Matches 250+	-1.8*** (0.3)	-1.1*** (0.2)	-0.4* (0.2)	1.9*** (0.4)	5.8*** (0.8)
Win %	0.02 (0.02)	0.01 (0.01)	0.00 (0.01)	-0.00 (0.01)	-0.01 (0.01)

Notes. Control function estimates using the generalized residual from P_i^{LL} . WTA non-Grand-Slam sample ($N = 2,545$), 2009–2024. All specifications control for pre-treatment ranking points (linear and quadratic), Elo / 100 (linear and quadratic), prior GS LL spots won, prior GS LL candidacies not won, prior non-GS LL spots won, prior non-GS LL candidacies not won, age, and event fixed effects. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 33: WTA Non-Grand-Slam: Heterogeneous Effects (Control Function)

	4w		8w		12w		26w		
	β_h	$\beta_h \times X_k$	β_h	$\beta_h \times X_k$	β_h	$\beta_h \times X_k$	β_h	$\beta_h \times X_k$	β_h
<i>Interaction: Above-Median Rank</i>									
Ranking Points Δ	11.42 (7.68)	24.60* (13.25)	9.92 (8.06)	19.37 (13.81)	11.78 (8.65)	17.72 (15.02)	7.93 (12.35)	-5.71 (21.53)	-25.1 (22.2)
Elo Δ	3.60 (3.45)	4.60 (3.99)	3.45 (3.69)	3.19 (4.24)	5.60 (4.10)	5.19 (4.62)	2.92 (5.94)	0.31 (6.68)	-2.6 (8.7)
Main Draws	0.71*** (0.12)	-2.23*** (0.17)	0.28** (0.11)	-0.95*** (0.16)	0.15 (0.11)	-0.18 (0.17)	-0.80*** (0.16)	3.17*** (0.24)	-2.71 (0.3)
Matches (250+)	1.10*** (0.22)	-3.99*** (0.37)	0.38* (0.20)	-1.91*** (0.35)	0.09 (0.20)	-0.49 (0.34)	-1.44*** (0.29)	5.16*** (0.48)	-4.79 (0.5)
Win %	0.03 (0.02)	-0.03 (0.03)	0.02 (0.02)	-0.02 (0.02)	0.01 (0.02)	-0.01 (0.02)	-0.00 (0.01)	-0.01 (0.02)	-0.0 (0.0)
<i>Interaction: Above-Median Age</i>									
Ranking Points Δ	17.48* (10.34)	16.82 (11.70)	21.09* (11.57)	0.14 (13.87)	31.61** (13.56)	-18.34 (15.99)	41.39* (21.27)	-78.14*** (27.06)	81.6 (44.9)
Elo Δ	3.92 (3.05)	5.05 (3.22)	2.26 (3.45)	6.12 (3.84)	6.96* (3.72)	3.90 (4.27)	6.80 (5.24)	-7.47 (6.13)	12.6 (7.1)
Main Draws	-0.99*** (0.16)	0.11 (0.21)	-0.69*** (0.14)	0.39** (0.17)	-0.30** (0.14)	0.40** (0.17)	1.03*** (0.23)	0.37 (0.29)	3.39* (0.5)
Matches (250+)	-1.87*** (0.33)	0.09 (0.40)	-1.28*** (0.30)	0.34 (0.33)	-0.50* (0.30)	0.14 (0.34)	2.05*** (0.49)	-0.40 (0.61)	6.92* (1.2)
Win %	0.00 (0.02)	0.03 (0.02)	-0.01 (0.02)	0.03* (0.02)	-0.01 (0.01)	0.03 (0.02)	-0.01 (0.01)	0.02 (0.01)	-0.0 (0.0)
<i>Interaction: Had Prior LL</i>									
Ranking Points Δ	22.52** (10.31)	7.80 (12.39)	23.99** (11.05)	-5.43 (14.53)	32.01** (13.06)	-19.87 (16.21)	27.07 (20.18)	-49.66* (26.40)	62.5 (42.5)
Elo Δ	5.13* (2.87)	2.92 (3.16)	6.00* (3.24)	-1.14 (3.57)	10.87*** (3.47)	-3.93 (4.24)	7.31 (5.39)	-8.37 (6.31)	6.6 (7.8)
Main Draws	-0.62*** (0.14)	-0.60*** (0.19)	-0.35*** (0.13)	-0.25 (0.16)	-0.12 (0.13)	0.08 (0.17)	0.89*** (0.21)	0.67** (0.27)	2.19* (0.5)
Matches (250+)	-1.27*** (0.28)	-1.08*** (0.38)	-0.75*** (0.26)	-0.69** (0.34)	-0.28 (0.27)	-0.27 (0.35)	1.66*** (0.44)	0.39 (0.58)	4.77* (1.1)
Win %	0.01 (0.02)	0.02 (0.02)	0.01 (0.02)	0.00 (0.02)	-0.00 (0.01)	0.02 (0.02)	-0.01 (0.01)	0.02 (0.01)	-0.0 (0.0)
Z^{pre} controls					Yes (main effects)				

Notes. Control function estimates by subgroup. WTA non-Grand-Slam sample, 2009–2024. Each row reports a set of estimates. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 34: WTA Non-Grand-Slam: First LL Episode Only (Control Function)

	4w	8w	12w	26w	52w
Ranking Points Δ	37.26** (14.46)	34.40** (15.07)	41.76** (16.99)	21.30 (22.60)	24.07 (42.09)
Elo Δ	13.25** (6.29)	14.30** (6.80)	17.18** (7.15)	5.88 (8.84)	-0.42 (11.73)
Main Draws	-1.13*** (0.27)	-0.60** (0.23)	-0.30 (0.22)	1.13*** (0.33)	2.53*** (0.73)
Matches (250+)	-2.30*** (0.51)	-1.34*** (0.43)	-0.90** (0.42)	1.42** (0.60)	4.33*** (1.41)
N	2,596				
Z^{pre} controls	Yes				

Notes. Control function estimates restricted to each player’s first LL candidacy episode. WTA non-Grand-Slam sample ($N = 1,423$, 378 treated), 2009–2024. Stacked specification with horizon-specific treatment effects. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B.3 Full Estimation Sample: Tournament Performance

Tables 35 and 36 report the match-level analysis of Section 6.3 with four-specification robustness. The ATP sample ($N = 8,543$ matches from 130 unique players) yields $\hat{\delta} = -0.123$ (bootstrap SE = 0.079, $p = 0.119$); the WTA sample ($N = 5,391$ matches from 98 unique players) yields $\hat{\delta} = +0.034$ (bootstrap SE = 0.112, $p = 0.760$). Tables 37 and 38 report four-specification robustness.

B.4 Non-Grand-Slam Tournament Performance Model

At non-Grand-Slam events, LL assignment is determined by ranking among qualifying losers rather than by lottery, so the treatment indicator D_{ie} is potentially endogenous. I use a control function approach: the generalized residual $\hat{v}_{ie}$, computed analytically from the selection probability P_i^{LL} (see Appendix A.4), is included in the outcome logit to absorb correlation between LL assignment and unobserved player quality. Standard errors are obtained via player-level block bootstrap (200 replications).

The ATP non-GS estimate is $\hat{\delta} = +0.033$ (bootstrap SE = 0.050, $p = 0.513$), based

Table 35: Grand Slam Tournament Performance: ATP (Full Sample)

	GS-ATP
$\hat{\delta}$ (LL entry)	-0.1227 (0.0787)
Matches	8,543
Players	130

Notes. Logit estimates of the effect of LL entry on match win probability at subsequent tournaments. Grand Slam ATP sample: 8,543 matches from 130 unique players. Sample includes all matches (main draw, qualifying, and challenger events) within each episode's observation window. $\hat{\delta}$ is the coefficient on the LL indicator in log-odds units. Bootstrap standard errors (200 replications) clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 36: Grand Slam Tournament Performance: WTA (Full Sample)

GS-WTA	
$\hat{\delta}$ (LL entry)	0.0342 (0.1118)
Matches	5,391
Players	98

Notes. Logit estimates of the effect of LL entry on match win probability at subsequent tournaments. Grand Slam WTA sample: 5,391 matches from 98 unique players. $\hat{\delta}$ is the coefficient on the LL indicator in log-odds units. Bootstrap standard errors (200 replications) clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 37: Grand Slam Tournament Robustness: ATP (Full Sample, Four Specifications)

Specification	N_{matches}	N_{players}	$\hat{\delta}$	SE	OR
Pooled	8,543	130	-0.1227	(0.0787)	0.885
Dose (matches won)	8,543	130	-0.1172**	(0.0492)	0.889

Notes. Four specifications for the full Grand Slam ATP sample. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 38: Grand Slam Tournament Robustness: WTA (Full Sample, Four Specifications)

Specification	N_{matches}	N_{players}	$\hat{\delta}$	SE	OR
Pooled	5,391	98	0.0342	(0.1118)	1.035
Dose (matches won)	5,391	98	0.0337	(0.0673)	1.034

Notes. Four specifications for the full Grand Slam WTA sample. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

on 113,189 matches from 838 unique players. The WTA non-GS estimate is $\hat{\delta} = -0.060$ (bootstrap SE = 0.032, $p = 0.061$), based on 80,168 matches from 747 unique players. The ATP estimate is small and statistically insignificant, consistent with no effect of LL entry on match-level performance. The WTA estimate is marginally significant and negative—an unexpected result that may reflect selection into tougher early-round opponents for ranking-based LL recipients. This finding does not alter the overall conclusion that match-level performance effects are generally null.

Tables 39 and 40 report the full non-Grand-Slam tournament performance results.

Table 39: Non-Grand-Slam Tournament Performance: ATP (Control Function)

Non-GS ATP	
$\hat{\delta}$ (LL entry)	0.0329 (0.0502)
$\hat{\rho}$ (CF)	-0.0327*** (0.0076)
Matches	113,189
Players	838

Notes. Control-function logit estimates of the effect of LL entry on match win probability at subsequent tournaments. Non-Grand-Slam ATP sample. $\hat{\delta}$ is the coefficient on the LL indicator in log-odds units. $\hat{\rho}$ is the coefficient on the control function residual. Standard errors from player-level block bootstrap (200 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B.5 First-LL Robustness: Grand Slam Tournament Performance

Tables 41 and 42 report four-specification robustness for the first-LL Grand Slam tournament performance model.

Table 40: Non-Grand-Slam Tournament Performance: WTA (Control Function)

Non-GS WTA	
$\hat{\delta}$ (LL entry)	-0.0599* (0.0320)
$\hat{\rho}$ (CF)	0.0147 (0.0097)
Matches	80,168
Players	747

Notes. Control-function logit estimates of the effect of LL entry on match win probability at subsequent tournaments. Non-Grand-Slam WTA sample. $\hat{\delta}$ is the coefficient on the LL indicator in log-odds units. $\hat{\rho}$ is the coefficient on the control function residual. Standard errors from player-level block bootstrap (200 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 41: Grand Slam Tournament Robustness: ATP (First-LL, Four Specifications)

Specification	N_{matches}	N_{players}	$\hat{\delta}$	SE	OR
Pooled	8,543	130	-0.1227	(0.0787)	0.885
Dose (matches won)	8,543	130	-0.1172**	(0.0492)	0.889

Notes. Four specifications for the Grand Slam ATP first-LL sample. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 42: Grand Slam Tournament Robustness: WTA (First-LL, Four Specifications)

Specification	N_{matches}	N_{players}	$\hat{\delta}$	SE	OR
Pooled	5,391	98	0.0342	(0.1118)	1.035
Dose (matches won)	5,391	98	0.0337	(0.0673)	1.034

Notes. Four specifications for the Grand Slam WTA first-LL sample. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B.6 Non-Grand-Slam Tournament Performance: First-LL Restriction

Tables 43 and 44 report the first-LL tournament performance model at non-Grand-Slam events, using the control function approach. The generalized residual $\hat{v}_{ie}$, computed analytically from P_i^{LL} , is included in the outcome logit because D_{ie} is endogenous at non-GS events (assigned by ranking rather than lottery).

Table 43: Non-Grand-Slam Tournament Performance: ATP (First-LL, Control Function)

	NonGS-ATP
$\hat{\delta}$ (LL entry)	0.0329 (0.0502)
$\hat{\rho}$ (CF)	-0.0327*** (0.0076)
Matches	113,189
Players	838

Notes. Control-function logit estimates restricted to first-time LL candidates. Non-Grand-Slam ATP sample. $\hat{\delta}$ is the coefficient on the LL indicator in log-odds units. $\hat{\rho}$ is the coefficient on the control function residual. Standard errors from player-level block bootstrap (200 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B.7 Fisher Randomization Inference

Table 45 reports Fisher randomization inference results.

Table 44: Non-Grand-Slam Tournament Performance: WTA (First-LL, Control Function)

NonGS-WTA	
$\hat{\delta}$ (LL entry)	-0.0599* (0.0320)
$\hat{\rho}$ (CF)	0.0147 (0.0097)
Matches	80,168
Players	747

Notes. Control-function logit estimates restricted to first-time LL candidates. Non-Grand-Slam WTA sample. $\hat{\delta}$ is the coefficient on the LL indicator in log-odds units. $\hat{\rho}$ is the coefficient on the control function residual. Standard errors from player-level block bootstrap (200 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B.8 Pre-Treatment Balance: Non-Grand-Slam

B.9 Full Dynamic Regression Tables

B.10 Consolidated Tournament Performance

B.11 Horizon and Dose Interaction Tables

B.12 Summary Impacts by Horizon

Table 45: Fisher Randomization Inference: Grand Slam Sample

	4w	8w	12w	26w	52w
<i>Panel A: ATP</i>					
Ranking pts Δ	29.897*** (12.852)	28.244* (13.468)	38.130** (14.316)	48.420* (24.727)	-3.166 (40.363)
Asymptotic p	0.020	0.036	0.008	0.050	0.937
Fisher p	0.007	0.072	0.036	0.095	0.958
Main draws entered	0.307* (0.258)	0.566** (0.252)	0.724** (0.256)	1.142*** (0.355)	0.881 (0.865)
Asymptotic p	0.234	0.025	0.005	0.001	0.309
Fisher p	0.071	0.022	0.017	0.009	0.408
Matches (250+)	0.385 (0.492)	0.755 (0.462)	0.946* (0.463)	1.723** (0.665)	1.362 (1.620)
Asymptotic p	0.435	0.102	0.041	0.010	0.400
Fisher p	0.260	0.114	0.090	0.038	0.440
Elo Δ	-2.184 (5.000)	-1.071 (5.999)	-4.705 (6.197)	4.395 (8.891)	-13.390 (12.434)
Asymptotic p	0.662	0.858	0.448	0.621	0.282
Fisher p	0.634	0.861	0.487	0.656	0.299
<i>Panel B: WTA</i>					
Ranking pts Δ	51.358*** (17.413)	63.657*** (18.776)	19.386 (21.289)	11.633 (38.427)	69.320 (60.985)
Asymptotic p	0.003	0.001	0.362	0.762	0.256
Fisher p	0.006	0.006	0.481	0.796	0.275
Main draws entered	0.167 (0.299)	0.418 (0.321)	0.378 (0.341)	0.790 (0.482)	1.056 (1.174)
Asymptotic p	0.576	0.192	0.268	0.101	0.369
Fisher p	0.383	0.105	0.270	0.145	0.347
Matches (250+)	0.421 (0.462)	0.992* (0.446)	0.829 (0.458)	1.566 (0.791)	0.913 (1.859)
Asymptotic p	0.363	0.026	0.070	0.048	0.623
Fisher p	0.275	0.056	0.105	0.103	0.615
Elo Δ	7.759 (5.204)	5.136 (6.871)	2.700 (8.372)	21.351* (10.196)	20.773 (15.516)
Asymptotic p	0.136	0.455	0.747	0.036	0.181
Fisher p	0.197	0.527	0.780	0.079	0.294
Permutations: 1,000; treatment permuted within slam $\times$ year					
N (ATP)		215 $\times$ 5 horizons			
N (WTA)		121 $\times$ 5 horizons			

Notes. Fisher p -values computed from 1,000 permutations within Grand Slam $\times$ Year blocks. ATP Grand Slam sample ($N = 248$). Asymptotic p -values from standard t -tests for comparison.

Table 46: Fisher Randomization Inference: Non-Grand-Slam Sample

	4w	8w	12w	26w	52w
<i>Panel A: ATP</i>					
Ranking pts Δ	12.585*** (6.588)	12.145** (6.934)	6.572 (7.816)	24.733 (10.666)	-7.994 (19.174)
Asymptotic p	0.056	0.080	0.400	0.020	0.677
Fisher p	0.009	0.036	0.283	0.140	0.730
Main draws entered	-1.400*** (0.132)	-0.900*** (0.123)	-0.326 (0.120)	1.543*** (0.190)	4.545*** (0.460)
Asymptotic p	0.000	0.000	0.006	0.000	0.000
Fisher p	0.000	0.001	0.359	0.000	0.000
Matches (250+)	-2.639*** (0.256)	-1.745*** (0.240)	-0.777 (0.240)	2.621*** (0.373)	7.866*** (0.878)
Asymptotic p	0.000	0.000	0.001	0.000	0.000
Fisher p	0.000	0.000	0.243	0.000	0.000
Elo Δ	0.150 (2.345)	0.322 (2.655)	-0.916 (2.870)	0.455 (3.891)	-5.208 (5.587)
Asymptotic p	0.949	0.903	0.750	0.907	0.351
Fisher p	0.937	0.898	0.713	0.898	0.404
<i>Panel B: WTA</i>					
Ranking pts Δ	27.444*** (9.419)	22.218** (10.135)	22.975* (11.300)	3.084 (15.613)	4.678 (31.035)
Asymptotic p	0.004	0.028	0.042	0.843	0.880
Fisher p	0.008	0.044	0.066	0.833	0.894
Main draws entered	-0.936*** (0.129)	-0.486** (0.114)	-0.087 (0.113)	1.219** (0.164)	3.155*** (0.414)
Asymptotic p	0.000	0.000	0.440	0.000	0.000
Fisher p	0.000	0.031	0.388	0.026	0.000
Matches (250+)	-1.812*** (0.263)	-1.092** (0.244)	-0.418 (0.241)	1.853 (0.356)	5.764*** (0.819)
Asymptotic p	0.000	0.000	0.083	0.000	0.000
Fisher p	0.000	0.020	0.160	0.122	0.002
Elo Δ	6.631* (2.527)	5.465 (2.756)	8.940 (2.942)	3.086 (4.359)	2.026 (5.944)
Asymptotic p	0.009	0.047	0.002	0.479	0.733
Fisher p	0.086	0.100	0.141	0.567	0.729
Permutations: 1,000; treatment permuted within tournament $\times$ year					
N (ATP)		2,096 $\times$ 5 horizons			
N (WTA)		1,928 $\times$ 5 horizons			

Notes. Treatment permuted within tournament $\times$ year blocks. Control function recomputed for each permutation. 1,000 permutations.

Table 47: Pre-Treatment Balance: Non-Grand-Slam Samples

Variable	ATP			WTA		
	LL	Control	p	LL	Control	p
Ranking points / 1000	0.51	0.36	0.000	0.64	0.45	0.000
(Ranking points / 1000) ²	0.32	0.18	0.000	0.51	0.28	0.000
Elo / 100	17.49	16.86	0.000	19.38	18.84	0.000
(Elo / 100) ²	307.01	285.52	0.000	376.82	356.51	0.000
Surface Elo / 100	17.01	16.67	0.000	18.46	18.19	0.000
(Surface Elo / 100) ²	291.02	279.31	0.000	342.69	332.76	0.000
Age	27.06	26.46	0.000	25.35	24.91	0.018
Had prior LL	0.54	0.43	0.000	0.50	0.42	0.000
Prior GS LL won	0.17	0.13	0.048	0.10	0.08	0.113
Prior GS LL not won	0.74	0.70	0.405	0.41	0.36	0.068
Prior non-GS LL won	0.96	0.79	0.004	0.96	0.70	0.000
Prior non-GS LL not won	1.66	1.37	0.000	1.78	1.65	0.117
Joint F -test p -value	0.000			0.000		
N	2708			2545		

Notes. Non-GS sample is not randomized; imbalance is expected and motivates the control function approach. ATP: $N = 2,708$; WTA: $N = 2,545$.

Table 48: ATP Grand Slam: Full Dynamic Effects Regression

	Ranking Points Δ	Elo Δ	Main Draws	Matches (250+)
<i>LL effect by horizon</i>				
LL $\times$ 4w	29.90** (12.85)	-2.18 (5.00)	0.31 (0.26)	0.38 (0.49)
LL $\times$ 8w	28.24** (13.47)	-1.07 (6.00)	0.57** (0.25)	0.76 (0.46)
LL $\times$ 12w	38.13*** (14.32)	-4.71 (6.20)	0.72*** (0.26)	0.95** (0.46)
LL $\times$ 26w	48.42* (24.73)	4.39 (8.89)	1.14*** (0.35)	1.72** (0.66)
LL $\times$ 52w	-3.17 (40.36)	-13.39 (12.43)	0.88 (0.87)	1.36 (1.62)
N	1,073			
Z^{pre} controls	Yes (interacted with horizon)			
Fixed effects	Slam $\times$ Year + Horizon			
Clustering	Player			

Notes. Full regression output from the stacked dynamic model. All Z_{ie}^{pre} covariates interacted with horizon. Event fixed effects and horizon fixed effects included. Standard errors clustered at player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 49: WTA Grand Slam: Full Dynamic Effects Regression

	Ranking Points Δ	Elo Δ	Main Draws	Matches (250+)
<i>LL effect by horizon</i>				
LL $\times$ 4w	51.36*** (17.41)	7.76 (5.20)	0.17 (0.30)	0.42 (0.46)
LL $\times$ 8w	63.66*** (18.78)	5.14 (6.87)	0.42 (0.32)	0.99** (0.45)
LL $\times$ 12w	19.39 (21.29)	2.70 (8.37)	0.38 (0.34)	0.83* (0.46)
LL $\times$ 26w	11.63 (38.43)	21.35** (10.20)	0.79 (0.48)	1.57* (0.79)
LL $\times$ 52w	69.32 (60.99)	20.77 (15.52)	1.06 (1.17)	0.91 (1.86)
N	605			
Z^{pre} controls	Yes (interacted with horizon)			
Fixed effects	Slam $\times$ Year + Horizon			
Clustering	Player			

Notes. Full regression output from the stacked dynamic model. All Z_{ie}^{pre} covariates interacted with horizon. Event fixed effects and horizon fixed effects included. Standard errors clustered at player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 50: ATP Non-Grand-Slam: Full Dynamic Effects Regression

	Ranking Points Δ	Elo Δ	Main Draws	Matches (250+)
<i>LL effect by horizon</i>				
LL $\times$ 4w	12.58* (6.59)	0.16 (2.34)	-1.40*** (0.13)	-2.64*** (0.26)
LL $\times$ 8w	12.14* (6.94)	0.34 (2.65)	-0.90*** (0.12)	-1.74*** (0.24)
LL $\times$ 12w	6.57 (7.82)	-0.91 (2.87)	-0.32*** (0.12)	-0.78*** (0.24)
LL $\times$ 26w	24.73** (10.67)	0.47 (3.89)	1.54*** (0.19)	2.62*** (0.37)
LL $\times$ 52w	-8.00 (19.18)	-5.21 (5.59)	4.55*** (0.46)	7.87*** (0.88)
<i>Selection correction ($\hat{\rho}_h$)</i>				
$\hat{v} \times$ 4w	5.96** (2.89)	0.87 (0.93)	0.53*** (0.05)	0.99*** (0.10)
$\hat{v} \times$ 8w	4.44 (3.02)	-0.15 (1.15)	0.37*** (0.05)	0.71*** (0.10)
$\hat{v} \times$ 12w	3.56 (2.97)	0.34 (1.28)	0.20*** (0.05)	0.41*** (0.09)
$\hat{v} \times$ 26w	-1.54 (4.77)	-0.56 (1.98)	-0.42*** (0.08)	-0.64*** (0.17)
$\hat{v} \times$ 52w	13.24 (9.08)	0.57 (2.82)	-1.24*** (0.19)	-2.12*** (0.38)
N	10,482			
Z^{pre} controls	Yes (main effects)			
Fixed effects	Tournament + Horizon			
Clustering	Player			

Notes. Full regression output from the stacked dynamic model with control function correction. All Z_{ie}^{pre} covariates interacted with horizon. Event fixed effects and horizon fixed effects included. Standard errors clustered at player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 51: WTA Non-Grand-Slam: Full Dynamic Effects Regression

	Ranking Points Δ	Elo Δ	Main Draws	Matches (250+)
<i>LL effect by horizon</i>				
LL $\times$ 4w	27.44*** (9.42)	6.63*** (2.53)	-0.94*** (0.13)	-1.81*** (0.26)
LL $\times$ 8w	22.22** (10.13)	5.47** (2.76)	-0.49*** (0.11)	-1.09*** (0.24)
LL $\times$ 12w	22.97** (11.30)	8.94*** (2.94)	-0.09 (0.11)	-0.42* (0.24)
LL $\times$ 26w	3.08 (15.61)	3.09 (4.36)	1.22*** (0.16)	1.85*** (0.36)
LL $\times$ 52w	4.68 (31.03)	2.03 (5.94)	3.15*** (0.41)	5.76*** (0.82)
<i>Selection correction ($\hat{\rho}_h$)</i>				
$\hat{v} \times$ 4w	-3.07 (3.36)	-1.77** (0.87)	0.33*** (0.05)	0.67*** (0.10)
$\hat{v} \times$ 8w	-0.62 (3.41)	-0.46 (0.98)	0.20*** (0.04)	0.45*** (0.09)
$\hat{v} \times$ 12w	-2.04 (3.73)	-1.36 (1.11)	0.07* (0.04)	0.23*** (0.09)
$\hat{v} \times$ 26w	-5.80 (5.91)	-1.87 (1.54)	-0.44*** (0.07)	-0.68*** (0.13)
$\hat{v} \times$ 52w	-6.15 (10.72)	-0.55 (2.69)	-1.13*** (0.17)	-2.06*** (0.32)
N	9,642			
Z^{pre} controls	Yes (main effects)			
Fixed effects	Tournament + Horizon			
Clustering	Player			

Notes. Full regression output from the stacked dynamic model with control function correction. All Z_{ie}^{pre} covariates interacted with horizon. Event fixed effects and horizon fixed effects included. Standard errors clustered at player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 52: Consolidated Tournament Performance: Grand Slam ATP

	(1) Pooled	(2) Dose: Wins	(3) Dose: Perf
D_{ie}	-0.1227** (0.0491)	-0.1555*** (0.0540)	-0.0517 (0.0523)
$D_{ie} \times$ Wins		0.0791 (0.0545)	
$D_{ie} \times$ Dose			0.0965*** (0.0240)
Matches	8,543	8,543	8,543

Notes. Three specifications: (1) pooled treatment effect, (2) dose by matches won (uncentered), (3) dose by performance probability ($-\text{logit}(\pi_{ie})$). Full covariate output. Bootstrap SEs (200 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 53: Consolidated Tournament Performance: Grand Slam WTA

	(1) Pooled	(2) Dose: Wins	(3) Dose: Perf
D_{ie}	0.0342 (0.0673)	0.0502 (0.0766)	0.0090 (0.0711)
$D_{ie} \times \text{Wins}$		-0.0461 (0.1051)	
$D_{ie} \times \text{Dose}$			-0.0437 (0.0397)
Matches	5,391	5,391	5,391

Notes. Three specifications: (1) pooled treatment effect, (2) dose by matches won (uncentered), (3) dose by performance probability ($-\logit(\pi_{ie})$). Full covariate output. Bootstrap SEs (200 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 54: Consolidated Tournament Performance: Non-Grand-Slam ATP

	(1) Pooled	(2) Dose: Wins	(3) Dose: Perf
D_{ie}	0.3834*** (0.0186)	0.4653*** (0.0201)	0.3498*** (0.0186)
$D_{ie} \times \text{Wins}$		-0.1620*** (0.0161)	
$D_{ie} \times \text{Dose}$			0.0581*** (0.0119)
Matches	458,789	458,789	458,789

Notes. Three specifications: (1) pooled treatment effect, (2) dose by matches won (uncentered), (3) dose by performance probability ($-\logit(\pi_{ie})$). Control function correction included. Bootstrap SEs (200 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 55: Consolidated Tournament Performance: Non-Grand-Slam WTA

	(1) Pooled	(2) Dose: Wins	(3) Dose: Perf
D_{ie}	-0.0563*** (0.0211)	-0.0340 (0.0236)	-0.0563*** (0.0211)
$D_{ie} \times \text{Wins}$		-0.0411* (0.0215)	
$D_{ie} \times \text{Dose}$			0.0044 (0.0138)
Matches	80,168	80,168	80,168

Notes. Three specifications: (1) pooled treatment effect, (2) dose by matches won (uncentered), (3) dose by performance probability ($-\logit(\pi_{ie})$). Control function correction included. Bootstrap SEs (200 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 56: Match-Level Horizon and Performance Dose Interaction: Grand Slam

	4w	8w	12w	26w	52w
<i>ATP</i>					
Total at dose= 0	-0.1175	-0.1109	-0.0886	-0.0078	-0.0424
Total at mean dose	-0.0919	-0.0592	0.0760	0.0675	0.0870
<i>WTA</i>					
Total at dose= 0	0.1466	-0.1041	-0.0057	-0.0739	0.0360
Total at mean dose	0.1345	0.0047	-0.1366	-0.0579	-0.0584
<i>Notes.</i> Combined ATP/WTA results. Spec 1: horizon-only treatment effects. Spec 2: horizon effects interacted with performance dose ($-\text{logit}(\pi_{ie})$). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.					

Table 57: Match-Level Horizon and Performance Dose Interaction: Non-Grand-Slam

	4w	8w	12w	26w	52w
<i>ATP</i>					
Total at dose= 0	0.0556	0.1096	-0.5647	0.4601	0.2051
Total at mean dose	0.0824	0.0626	-0.5480	0.4525	0.2321
<i>WTA</i>					
Total at dose= 0	-0.1682	-0.1974	-0.0241	-0.0677	0.0312
Total at mean dose	-0.1132	-0.1993	0.0495	-0.0529	-0.0047
<i>Notes.</i> Combined ATP/WTA results with control function correction. Spec 1: horizon-only treatment effects. Spec 2: horizon effects interacted with performance dose ($-\text{logit}(\pi_{ie})$). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.					

Table 58: Summary Match-Level Impacts by Horizon: Grand Slam

	ATP		WTA	
	Base	At mean dose	Base	At mean dose
<i>AME on match win probability</i>				
4w	-0.0295 [-0.0845, 0.0120]	-0.0291 [-0.0847, 0.0146]	0.0290 [-0.0351, 0.0893]	0.0306 [-0.0517, 0.1008]
8w	-0.0322 [-0.0842, 0.0243]	-0.0331 [-0.0897, 0.0252]	-0.0332 [-0.1203, 0.0386]	-0.0324 [-0.1311, 0.0365]
12w	-0.0439 [-0.1156, 0.0257]	-0.0471 [-0.1272, 0.0219]	0.0151 [-0.0664, 0.0853]	0.0139 [-0.0824, 0.0838]
26w	-0.0144 [-0.0518, 0.0316]	-0.0130 [-0.0568, 0.0315]	-0.0154 [-0.0657, 0.0419]	-0.0150 [-0.0700, 0.0417]
52w	-0.0281 [-0.0609, 0.0152]	-0.0261 [-0.0574, 0.0134]	0.0162 [-0.0432, 0.0596]	0.0158 [-0.0418, 0.0591]
<i>Cumulative expected ranking points</i>				
4w	-0.4	-0.4	0.4	0.5
8w	-0.9	-0.9	-0.1	-0.0
12w	-1.6	-1.6	0.2	0.2
26w	-1.8	-1.8	-0.1	-0.0
52w	-2.2	-2.2	0.2	0.2
<i>Cumulative expected additional wins</i>				
4w	-0.0295	-0.0291	0.0290	0.0306
8w	-0.0617	-0.0622	-0.0042	-0.0018
12w	-0.1055	-0.1093	0.0109	0.0122
26w	-0.1200	-0.1223	-0.0044	-0.0028
52w	-0.1481	-0.1484	0.0117	0.0130

Notes. Average marginal effects on match win probability, cumulative ranking points, and cumulative wins derived from the horizon model via delta method. 95% CIs from bootstrap (200 replications).

Table 59: Summary Match-Level Impacts by Horizon: Non-Grand-Slam

	ATP		WTA	
	Base	At mean dose	Base	At mean dose
<i>AME on match win probability</i>				
4w	-0.0125 [-0.0403, 0.0100]	-0.0156 [-0.0435, 0.0053]	-0.0350 [-0.0551, -0.0161]	-0.0338 [-0.0521, -0.0140]
8w	-0.0001 [-0.0257, 0.0197]	0.0004 [-0.0275, 0.0192]	-0.0413 [-0.0631, -0.0189]	-0.0413 [-0.0631, -0.0168]
12w	-0.0325 [-0.0756, -0.0106]	-0.0303 [-0.0654, -0.0101]	-0.0052 [-0.0273, 0.0161]	-0.0040 [-0.0268, 0.0187]
26w	0.0267 [-0.0160, 0.0547]	0.0172 [-0.0179, 0.0387]	-0.0143 [-0.0318, 0.0018]	-0.0139 [-0.0315, 0.0025]
52w	0.0086 [-0.0151, 0.0239]	0.0070 [-0.0147, 0.0224]	0.0064 [-0.0130, 0.0218]	0.0061 [-0.0127, 0.0213]
<i>Cumulative expected ranking points</i>				
4w	-0.1	-0.1	-0.2	-0.2
8w	-0.1	-0.1	-0.4	-0.4
12w	-0.2	-0.2	-0.4	-0.4
26w	-0.1	-0.1	-0.5	-0.5
52w	-0.0	-0.1	-0.4	-0.4
<i>Cumulative expected additional wins</i>				
4w	-0.0125	-0.0156	-0.0350	-0.0338
8w	-0.0126	-0.0152	-0.0762	-0.0751
12w	-0.0450	-0.0454	-0.0814	-0.0791
26w	-0.0184	-0.0282	-0.0957	-0.0931
52w	-0.0098	-0.0212	-0.0893	-0.0869

Notes. Average marginal effects on match win probability, cumulative ranking points, and cumulative wins derived from the horizon model via delta method. Control function correction included. 95% CIs from bootstrap (200 replications).